2.35 Integration by Substitution

Worked Solutions

1 a)

$\int (3x+1)^4 \, \mathrm{d}x$

$u = 3x+1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 3 \Rightarrow \mathrm{d}x = \frac{1}{3}\mathrm{d}u$

$\int (3x+1)^4 \, \mathrm{d}x = \int u^4 \times \frac{1}{3} \, \mathrm{d}u = \int \frac{1}{3}u^4 \, \mathrm{d}u$

$= \frac{1}{15}u^5 + c$

$= \frac{1}{15}(3x+1)^5 + c$

1 b)

$\int \frac{4}{(2x-3)^3} \, \mathrm{d}x$

$u = 2x-3 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2 \Rightarrow \mathrm{d}x = \frac{1}{2}\mathrm{d}u$

$\int \frac{4}{(2x-3)^3} \, \mathrm{d}x = \int \frac{4}{u^3} \times \frac{1}{2} \, \mathrm{d}u = \int 2u^{-3} \, \mathrm{d}u$

$= -u^{-2} + c$

$= -(2x-3)^{-2} + c = -\frac{1}{(2x-3)^2} + c$

1 c)

$\int \sqrt{8x+5} \, \mathrm{d}x$

$u = 8x+5 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 8 \Rightarrow \mathrm{d}x = \frac{1}{8}\mathrm{d}u$

$\int \sqrt{8x+5} \, \mathrm{d}x = \int \sqrt{u} \times \frac{1}{8} \, \mathrm{d}u = \int \frac{1}{8}u^{\frac{1}{2}} \, \mathrm{d}u$

$= \frac{1}{12}u^{\frac{3}{2}} + c$

$= \frac{1}{12}(8x+5)^{\frac{3}{2}} + c$

1 d)

$\int \frac{6}{\sqrt[3]{5-2x}} \, \mathrm{d}x$

$u = 5-2x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -2 \Rightarrow \mathrm{d}x = -\frac{1}{2}\mathrm{d}u$

$\int \frac{6}{\sqrt[3]{5-2x}} \, \mathrm{d}x = \int \frac{6}{u^{\frac{1}{3}}} \times -\frac{1}{2} \, \mathrm{d}u = \int -3u^{-\frac{1}{3}} \, \mathrm{d}u$

$= -\frac{9}{2}u^{\frac{2}{3}} + c$

$= -\frac{9}{2}(5-2x)^{\frac{2}{3}} + c$

2 a)

Translation by the vector $\begin{pmatrix} -3 \\ 0 \end{pmatrix}$

2 b)

$\int x(x-3)^5 \, \mathrm{d}x$

$u = x-3 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = \mathrm{d}u$

$u = x-3 \Rightarrow x = u+3$

$\int x(x-3)^5 \, \mathrm{d}x = \int (u+3)u^5 \, \mathrm{d}u$

3 a)

Translation by the vector $\begin{pmatrix} 4 \\ 0 \end{pmatrix}$

3 b)

$\int 2x\sqrt{x+4}\,\mathrm{d}x$

$u = x + 4 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = \mathrm{d}u$

$u = x + 4 \Rightarrow x = u - 4$

$\int 2x\sqrt{x+4}\,\mathrm{d}x = \int 2(u-4)\sqrt{u}\,\mathrm{d}u$

4 a)

$\int x\sqrt{x+1}\,\mathrm{d}x$

$u = x + 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = \mathrm{d}u$

$u = x + 1 \Rightarrow x = u - 1$

$\int x\sqrt{x+1}\,\mathrm{d}x = \int (u-1)u^{\frac{1}{2}}\,\mathrm{d}u = \int (u^{\frac{3}{2}} - u^{\frac{1}{2}})\,\mathrm{d}u$

$= \frac{2}{5}u^{\frac{5}{2}} - \frac{2}{3}u^{\frac{3}{2}} + c$

$= \frac{2}{5}(x+1)^{\frac{5}{2}} - \frac{2}{3}(x+1)^{\frac{3}{2}} + c$

4 b)

$\int 2x\sqrt{2x+3}\,\mathrm{d}x$

$u = 2x + 3 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2 \Rightarrow \mathrm{d}x = \frac{1}{2}\mathrm{d}u$

$u = 2x + 3 \Rightarrow 2x = u - 3$

$\int 2x\sqrt{2x+3}\,\mathrm{d}x = \int (u-3)u^{\frac{1}{2}} \times \frac{1}{2}\,\mathrm{d}u$

$= \int \left(\frac{1}{2}u^{\frac{3}{2}} - \frac{3}{2}u^{\frac{1}{2}}\right)\,\mathrm{d}u$

$= \frac{1}{5}u^{\frac{5}{2}} - u^{\frac{3}{2}} + c$

$= \frac{1}{5}(2x+3)^{\frac{5}{2}} - (2x+3)^{\frac{3}{2}} + c$

4 c)

$\int x^3\sqrt{x^2+1}\,\mathrm{d}x$

$u = x^2 + 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$u = x^2 + 1 \Rightarrow x^2 = u - 1$

$\int x^3\sqrt{x^2+1}\,\mathrm{d}x = \int x^3 u^{\frac{1}{2}} \times \frac{1}{2x}\,\mathrm{d}u = \int \frac{1}{2}x^2 u^{\frac{1}{2}}\,\mathrm{d}u$

$= \int \frac{1}{2}(u-1)u^{\frac{1}{2}}\,\mathrm{d}u = \int \left(\frac{1}{2}u^{\frac{3}{2}} - \frac{1}{2}u^{\frac{1}{2}}\right)\,\mathrm{d}u$

$= \frac{1}{5}u^{\frac{5}{2}} - \frac{1}{3}u^{\frac{3}{2}} + c$

$= \frac{1}{5}(x^2+1)^{\frac{5}{2}} - \frac{1}{3}(x^2+1)^{\frac{3}{2}} + c$

5 a)

$\int \frac{x}{x+1}\,\mathrm{d}x$

$u = x + 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = \mathrm{d}u$

$u = x + 1 \Rightarrow x = u - 1$

$\int \frac{x}{x+1}\,\mathrm{d}x = \int \frac{u-1}{u}\,\mathrm{d}u = \int \left(\frac{u}{u} - \frac{1}{u}\right)\,\mathrm{d}u = \int \left(1 - \frac{1}{u}\right)\,\mathrm{d}u$

$= u - \ln|u| + c$

$= (x+1) - \ln|x+1| + c$

5 b)

$\int \frac{2x}{2x+3}\,\mathrm{d}x$

$u = 2x+3 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2 \Rightarrow \mathrm{d}x = \frac{1}{2}\mathrm{d}u$

$u = 2x+3 \Rightarrow 2x = u-3$

$\int \frac{2x}{2x+3}\,\mathrm{d}x = \int \frac{u-3}{u} \times \frac{1}{2}\,\mathrm{d}u = \int \left(\frac{u}{2u} - \frac{3}{2u}\right)\,\mathrm{d}u$

$= \int \left(\frac{1}{2} - \frac{3}{2u}\right)\,\mathrm{d}u$

$= \frac{1}{2}u - \frac{3}{2}\ln|u| + c$

$= \frac{1}{2}(2x+3) - \frac{3}{2}\ln|2x+3| + c$

Alternative Method:

$= \frac{1}{2}u - \frac{3}{2}\ln|2u| + c$

$= \frac{1}{2}(2x+3) - \frac{3}{2}\ln|4x+6| + c$

5 c)

$\int \frac{x^3}{x^2+1}\,\mathrm{d}x$

$u = x^2+1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$u = x^2+1 \Rightarrow x^2 = u-1$

$\int \frac{x^3}{x^2+1}\,\mathrm{d}x = \int \frac{x^3}{u} \times \frac{1}{2x}\,\mathrm{d}u = \int \frac{x^2}{2u}\,\mathrm{d}u = \int \frac{u-1}{2u}\,\mathrm{d}u$

$= \int \left(\frac{u}{2u} - \frac{1}{2u}\right)\,\mathrm{d}u = \int \left(\frac{1}{2} - \frac{1}{2} \times \frac{1}{u}\right)\,\mathrm{d}u$

$= \frac{1}{2}u - \frac{1}{2}\ln|u| + c$

$= \frac{1}{2}(x^2+1) - \frac{1}{2}\ln|x^2+1| + c$

Alternative Method:

$= \frac{1}{2}u - \frac{1}{2}\ln|2u| + c$

$= \frac{1}{2}(x^2+1) - \frac{1}{2}\ln|2x^2+2| + c$

6 a)

$\int \frac{x^2}{x+1}\,\mathrm{d}x$

$u = x+1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = \mathrm{d}u$

$u = x+1 \Rightarrow x = u-1$

$\int \frac{x^2}{x+1}\,\mathrm{d}x = \int \frac{(u-1)^2}{u}\,\mathrm{d}u = \int \frac{u^2-2u+1}{u}\,\mathrm{d}u$

$= \int \left(\frac{u^2}{u} - \frac{2u}{u} + \frac{1}{u}\right)\,\mathrm{d}u = \int \left(u - 2 + \frac{1}{u}\right)\,\mathrm{d}u$

$= \frac{1}{2}u^2 - 2u + \ln|u| + c$

$= \frac{1}{2}(x+1)^2 - 2(x+1) + \ln|x+1| + c$

6 b)

$\int \frac{4x^2}{2x+3}\,\mathrm{d}x$

$u = 2x+3 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2 \Rightarrow \mathrm{d}x = \frac{1}{2}\mathrm{d}u$

$u = 2x+3 \Rightarrow 2x = u-3$

$\int \frac{4x^2}{2x+3}\,\mathrm{d}x = \int \frac{(u-3)^2}{u} \times \frac{1}{2}\,\mathrm{d}u = \int \frac{u^2-6u+9}{2u}\,\mathrm{d}u$

$= \int \left(\frac{u^2}{2u} - \frac{6u}{2u} + \frac{9}{2u}\right)\,\mathrm{d}u = \int \left(\frac{1}{2}u - 3 + \frac{9}{2u}\right)\,\mathrm{d}u$

$= \frac{1}{4}u^2 - 3u + \frac{9}{2}\ln|u| + c$

$= \frac{1}{4}(2x+3)^2 - 3(2x+3) + \frac{9}{2}\ln|2x+3| + c$

Alternative Method:

$= \frac{1}{4}u^2 - 3u + \frac{9}{2}\ln|2u| + c$

$= \frac{1}{4}(2x+3)^2 - 3(2x+3) + \frac{9}{2}\ln|4x+6| + c$

6 c)

$\int \frac{x^5}{x^2+1}\,\mathrm{d}x$

$u = x^2+1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$u = x^2+1 \Rightarrow x^2 = u-1$

$\int \frac{x^5}{x^2+1}\,\mathrm{d}x = \int \frac{x^5}{u} \times \frac{1}{2x}\,\mathrm{d}u = \int \frac{x^4}{2u}\,\mathrm{d}u = \int \frac{(u-1)^2}{2u}\,\mathrm{d}u$

$= \int \left(\frac{u^2}{2u} - \frac{2u}{2u} + \frac{1}{2u}\right)\,\mathrm{d}u = \int \left(\frac{1}{2}u - 1 + \frac{1}{2u}\right)\,\mathrm{d}u$

$= \frac{1}{4}u^2 - u + \frac{1}{2}\ln|u| + c$

$= \frac{1}{4}(x^2+1)^2 - (x^2+1) + \frac{1}{2}\ln|x^2+1| + c$

$= \frac{1}{4}x^4 - \frac{1}{2}x^2 + \frac{1}{2}\ln|x^2+1| + c$

7 a)

$\int x^2\sqrt{x+1}\,\mathrm{d}x$

$u = x+1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = \mathrm{d}u$

$u = x+1 \Rightarrow x = u-1$

$\int x^2\sqrt{x+1}\,\mathrm{d}x = \int (u-1)^2 u^{\frac{1}{2}}\,\mathrm{d}u$

$= \int (u^2 - 2u + 1)u^{\frac{1}{2}}\,\mathrm{d}u = \int (u^{\frac{5}{2}} - 2u^{\frac{3}{2}} + u^{\frac{1}{2}})\,\mathrm{d}u$

$= \frac{2}{7}u^{\frac{7}{2}} - \frac{4}{5}u^{\frac{5}{2}} + \frac{2}{3}u^{\frac{3}{2}} + c$

$= \frac{2}{7}(x+1)^{\frac{7}{2}} - \frac{4}{5}(x+1)^{\frac{5}{2}} + \frac{2}{3}(x+1)^{\frac{3}{2}} + c$

7 b)

$\int 4x^2\sqrt{2x+3}\,\mathrm{d}x$

$u = 2x+3 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2 \Rightarrow \mathrm{d}x = \frac{1}{2}\mathrm{d}u$

$u = 2x+3 \Rightarrow 2x = u-3$

$\int 4x^2\sqrt{2x+3}\,\mathrm{d}x = \int (u-3)^2 u^{\frac{1}{2}} \times \frac{1}{2}\,\mathrm{d}u$

$= \int \frac{1}{2}(u^2 - 6u + 9)u^{\frac{1}{2}}\,\mathrm{d}u = \int \left(\frac{1}{2}u^{\frac{5}{2}} - 3u^{\frac{3}{2}} + \frac{9}{2}u^{\frac{1}{2}}\right)\,\mathrm{d}u$

$= \frac{1}{7}u^{\frac{7}{2}} - \frac{6}{5}u^{\frac{5}{2}} + 3u^{\frac{3}{2}} + c$

$= \frac{1}{7}(2x+3)^{\frac{7}{2}} - \frac{6}{5}(2x+3)^{\frac{5}{2}} + 3(2x+3)^{\frac{3}{2}} + c$

7 c)

$\int x^3\sqrt{x^2+1}\,\mathrm{d}x$

$u = x^2+1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$u = x^2+1 \Rightarrow x^2 = u-1$

$\int x^3\sqrt{x^2+1}\,\mathrm{d}x = \int x^3\sqrt{u} \times \frac{1}{2x}\,\mathrm{d}u = \int \frac{1}{2}x^2u^{\frac{1}{2}}\,\mathrm{d}u$

$= \int \frac{1}{2}(u-1)u^{\frac{1}{2}}\,\mathrm{d}u = \int \left(\frac{1}{2}u^{\frac{3}{2}} - \frac{1}{2}u^{\frac{1}{2}}\right)\mathrm{d}u$

$= \frac{1}{5}u^{\frac{5}{2}} - \frac{1}{3}u^{\frac{3}{2}} + c$

$= \frac{1}{5}(x^2+1)^{\frac{5}{2}} - \frac{1}{3}(x^2+1)^{\frac{3}{2}} + c$

8)

$\int 189x(2-3x)^5\,\mathrm{d}x$

$u = 2-3x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -3 \Rightarrow \mathrm{d}x = -\frac{1}{3}\mathrm{d}u$

$u = 2-3x \Rightarrow x = \frac{2-u}{3}$

$\int 189x(2-3x)^5\,\mathrm{d}x = \int 189\left(\frac{2-u}{3}\right)u^5 \times -\frac{1}{3}\,\mathrm{d}u$

$= \int -21(2-u)u^5\,\mathrm{d}u = \int(-42u^5 + 21u^6)\,\mathrm{d}u$

$= -7u^6 + 3u^7 + c$

$= -7(2-3x)^6 + 3(2-3x)^7 + c$

9)

$\int_0^1 3x(x+1)^{-\frac{1}{2}}\,\mathrm{d}x$

$u = x+1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = \mathrm{d}u$

$u = x+1 \Rightarrow x = u-1$

Change limits:

$x = 1 \Rightarrow u = 1+1 = 2$

$x = 0 \Rightarrow u = 0+1 = 1$

$\int_0^1 3x(x+1)^{-\frac{1}{2}}\,\mathrm{d}x = \int_1^2 3(u-1)u^{-\frac{1}{2}}\,\mathrm{d}u$

$= \int_1^2 (3u^{\frac{1}{2}} - 3u^{-\frac{1}{2}})\,\mathrm{d}u$

$= \left[2u^{\frac{3}{2}} - 6u^{\frac{1}{2}}\right]_1^2$

$= \left(2\times 2^{\frac{3}{2}} - 6\times 2^{\frac{1}{2}}\right) - \left(2\times 1^{\frac{3}{2}} - 6\times 1^{\frac{1}{2}}\right)$

$= (4\sqrt{2} - 6\sqrt{2}) - (2-6) = (-2\sqrt{2}) - (-4)$

$= 4 - 2\sqrt{2}$

10)

$\int_{-2}^{-\frac{4}{3}} (6x+1)(3x+4)^7\,\mathrm{d}x$

$u = 3x+4 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 3 \Rightarrow \mathrm{d}x = \frac{1}{3}\mathrm{d}u$

$u = 3x+4 \Rightarrow 3x = u-4 \Rightarrow 6x = 2u-8$

$\Rightarrow 6x+1=2u-7$

Change limits:

$x=-\frac{4}{3} \Rightarrow u=3\left(-\frac{4}{3}\right)+4=0$

$x=-2 \Rightarrow u=3(-2)+4=-2$

$\int_{-2}^{-\frac{4}{3}}(6x+1)(3x+4)^7\,\mathrm{d}x=\int_{-2}^{0}(2u-7)u^7\times\frac{1}{3}\,\mathrm{d}u$

$=\int_{-2}^{0}\left(\frac{2}{3}u^8-\frac{7}{3}u^7\right)\,\mathrm{d}u$

$=\left[\frac{2}{27}u^9-\frac{7}{24}u^8\right]_{-2}^{0}$

$=\left(\frac{2}{27}\times 0^9-\frac{7}{24}\times 0^8\right)-\left(\frac{2}{27}\times(-2)^9-\frac{7}{24}\times(-2)^8\right)$

$=(0)-\left(-\frac{1024}{27}-\frac{224}{3}\right)=(0)-\left(-\frac{3040}{27}\right)=\frac{3040}{27}$

11)

$\int_1^2\frac{x}{x+2}\,\mathrm{d}x$

$u=x+2 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x}=1 \Rightarrow \mathrm{d}x=\mathrm{d}u$

$u=x+2 \Rightarrow x=u-2$

Change limits:

$x=2 \Rightarrow u=2+2=4$

$x=1 \Rightarrow u=1+2=3$

$\int_1^2\frac{x}{x+2}\,\mathrm{d}x=\int_3^4\frac{u-2}{u}\,\mathrm{d}u$

$=\int_3^4\left(\frac{u}{u}-\frac{2}{u}\right)\,\mathrm{d}u=\int_3^4\left(1-\frac{2}{u}\right)\,\mathrm{d}u$

$=[u-2\ln|u|]_3^4$

$=(4-2\ln 4)-(3-2\ln 3)$

$=1+2\ln 3-2\ln 4=1+2(\ln 3-\ln 4)$

$=1+2\ln\frac{3}{4}=1+\ln\left(\frac{3}{4}\right)^2=1+\ln\frac{9}{16}$

12)

$\int_1^3\frac{9x}{3x+5}\,\mathrm{d}x$

$u=3x+5 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x}=3 \Rightarrow \mathrm{d}x=\frac{1}{3}\mathrm{d}u$

$u=3x+5 \Rightarrow 3x=u-5 \Rightarrow 9x=3u-15$

Change limits:

$x=3 \Rightarrow u=3\times 3+5=14$

$x=1 \Rightarrow u=3\times 1+5=8$

$\int_1^3\frac{9x}{3x+5}\,\mathrm{d}x=\int_8^{14}\frac{3u-15}{u}\times\frac{1}{3}\,\mathrm{d}u$

$=\int_8^{14}\left(\frac{u}{u}-\frac{5}{u}\right)\,\mathrm{d}u=\int_8^{14}\left(1-\frac{5}{u}\right)\,\mathrm{d}u$

$=[u-5\ln|u|]_8^{14}$

$=(14-5\ln 14)-(8-5\ln 8)$

$=6+5\ln 8-5\ln 14=6+5(\ln 8-\ln 14)$

$=6+5\ln\frac{8}{14}=6+5\ln\frac{4}{7}$

13)

$\int_1^2 \frac{8x-1}{(2x+3)^3}\,\mathrm{d}x$

$u = 2x+3 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2 \Rightarrow \mathrm{d}x = \frac{1}{2}\mathrm{d}u$

$u = 2x+3 \Rightarrow 2x = u-3 \Rightarrow 8x = 4u-12$

$\Rightarrow 8x-1 = 4u-13$

Change limits:

$x = 2 \Rightarrow u = 2\times 2+3 = 7$

$x = 1 \Rightarrow u = 2\times 1+3 = 5$

$\int_1^2 \frac{8x-1}{(2x+3)^3}\,\mathrm{d}x = \int_5^7 \frac{4u-13}{u^3}\times\frac{1}{2}\,\mathrm{d}u$

$= \int_5^7 \left(\frac{4u}{2u^3} - \frac{13}{2u^3}\right)\,\mathrm{d}u = \int_5^7 \left(2u^{-2} - \frac{13}{2}u^{-3}\right)\,\mathrm{d}u$

$= \left[-2u^{-1} + \frac{13}{4}u^{-2}\right]_5^7$

$= \left(-2\times 7^{-1} + \frac{13}{4}\times 7^{-2}\right) - \left(-2\times 5^{-1} + \frac{13}{4}\times 5^{-2}\right)$

$= \left(-\frac{43}{196}\right) - \left(-\frac{27}{100}\right) = \frac{62}{1225}$

14)

$\int_{\frac{3}{2}}^{\frac{19}{2}} 5x\sqrt{2x-3}\,\mathrm{d}x$

$u = 2x-3 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2 \Rightarrow \mathrm{d}x = \frac{1}{2}\mathrm{d}u$

$u = 2x-3 \Rightarrow x = \frac{u+3}{2}$

Change limits:

$x = \frac{19}{2} \Rightarrow u = 2\times\frac{19}{2}-3 = 16$

$x = \frac{3}{2} \Rightarrow u = 2\times\frac{3}{2}-3 = 0$

$\int_{\frac{3}{2}}^{\frac{19}{2}} 5x\sqrt{2x-3}\,\mathrm{d}x = \int_0^{16} 5\left(\frac{u+3}{2}\right)u^{\frac{1}{2}}\times\frac{1}{2}\,\mathrm{d}u$

$= \int_0^{16} \frac{5}{4}(u+3)u^{\frac{1}{2}}\,\mathrm{d}u = \int_0^{16}\left(\frac{5}{4}u^{\frac{3}{2}} + \frac{15}{4}u^{\frac{1}{2}}\right)\,\mathrm{d}u$

$= \left[\frac{1}{2}u^{\frac{5}{2}} + \frac{5}{2}u^{\frac{3}{2}}\right]_0^{16}$

$= \left(\frac{1}{2}\times 16^{\frac{5}{2}} + \frac{5}{2}\times 16^{\frac{3}{2}}\right) - \left(\frac{1}{2}\times 0^{\frac{5}{2}} + \frac{5}{2}\times 0^{\frac{3}{2}}\right)$

$= (512+160) - (0+0) = 672$

15)

$\int_0^1 140x(4x+1)^{\frac{3}{2}}\,\mathrm{d}x$

$u = 4x+1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 4 \Rightarrow \mathrm{d}x = \frac{1}{4}\mathrm{d}u$

$u = 4x+1 \Rightarrow x = \frac{u-1}{4}$

Change limits:

$x = 1 \Rightarrow u = 4\times 1+1 = 5$

$x = 0 \Rightarrow u = 4\times 0+1 = 1$

$\int_0^1 140x(4x+1)^{\frac{3}{2}}\,\mathrm{d}x = \int_1^5 140\left(\frac{u-1}{4}\right)u^{\frac{3}{2}}\times\frac{1}{4}\,\mathrm{d}u$

$= \int_1^5 \frac{35}{4}(u-1)u^{\frac{3}{2}}\,\mathrm{d}u = \int_1^5\left(\frac{35}{4}u^{\frac{5}{2}} - \frac{35}{4}u^{\frac{3}{2}}\right)\,\mathrm{d}u$

$= \left[\frac{5}{2}u^{\frac{7}{2}} - \frac{7}{2}u^{\frac{5}{2}}\right]_1^5$

$= \left(\frac{5}{2} \times 5^{\frac{7}{2}} - \frac{7}{2} \times 5^{\frac{5}{2}}\right) - \left(\frac{5}{2} \times 1^{\frac{7}{2}} - \frac{7}{2} \times 1^{\frac{5}{2}}\right)$

$= \left(\frac{625}{2}\sqrt{5} - \frac{175}{2}\sqrt{5}\right) - \left(\frac{5}{2} - \frac{7}{2}\right)$

$= (225\sqrt{5}) - (-1) = 1 + 225\sqrt{5}$

16)

$\int (30x - 45)\sqrt{x+2}\,\mathrm{d}x$

$u = x + 2 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 1 \Rightarrow \mathrm{d}x = \mathrm{d}u$

$u = x + 2 \Rightarrow x = u - 2$

$\Rightarrow 30x - 45 = 30(u - 2) - 45$

$\Rightarrow 30x - 45 = 30u - 105$

$\int (30x - 45)\sqrt{x+2}\,\mathrm{d}x = \int (30u - 105)u^{\frac{1}{2}}\,\mathrm{d}u$

$= \int (30u^{\frac{3}{2}} - 105u^{\frac{1}{2}})\,\mathrm{d}u$

$= 12u^{\frac{5}{2}} - 70u^{\frac{3}{2}} + c$

$= 12(x+2)^{\frac{5}{2}} - 70(x+2)^{\frac{3}{2}} + c$

$= (x+2)^{\frac{3}{2}}(12(x+2) - 70) + c$

$= (x+2)^{\frac{3}{2}}(12x - 46) + c$

17)

$\int_{\frac{1}{12}}^{\frac{3}{4}} (2 - 6x)(12x - 1)^{\frac{1}{3}}\,\mathrm{d}x$

$u = 12x - 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 12 \Rightarrow \mathrm{d}x = \frac{1}{12}\mathrm{d}u$

$u = 12x - 1 \Rightarrow x = \frac{u+1}{12} \Rightarrow 2 - 6x = 2 - 6\left(\frac{u+1}{12}\right)$

$\Rightarrow 2 - 6x = 2 - \frac{1}{2}(u + 1) \Rightarrow 2 - 6x = \frac{3}{2} - \frac{1}{2}u$

Change limits:

$x = \frac{3}{4} \Rightarrow u = 12 \times \frac{3}{4} - 1 = 8$

$x = \frac{1}{12} \Rightarrow u = 12 \times \frac{1}{12} - 1 = 0$

$\int_{\frac{1}{12}}^{\frac{3}{4}} (2 - 6x)(12x - 1)^{\frac{1}{3}}\,\mathrm{d}x = \int_0^8 \left(\frac{3}{2} - \frac{1}{2}u\right) u^{\frac{1}{3}} \times \frac{1}{12}\,\mathrm{d}u$

$= \int_0^8 \left(\frac{1}{8} - \frac{1}{24}u\right) u^{\frac{1}{3}}\,\mathrm{d}u = \int_0^8 \left(\frac{1}{8}u^{\frac{1}{3}} - \frac{1}{24}u^{\frac{4}{3}}\right)\,\mathrm{d}u$

$= \left[\frac{3}{32}u^{\frac{4}{3}} - \frac{1}{56}u^{\frac{7}{3}}\right]_0^8$

$= \left(\frac{3}{32} \times 8^{\frac{4}{3}} - \frac{1}{56} \times 8^{\frac{7}{3}}\right) - \left(\frac{3}{32} \times 0^{\frac{4}{3}} - \frac{1}{56} \times 0^{\frac{7}{3}}\right)$

$= \left(\frac{3}{2} - \frac{16}{7}\right) - (0 - 0) = -\frac{11}{14}$

18)

$\int \frac{2x^3 - x}{\sqrt{x^2+5}}\,\mathrm{d}x$

$u = x^2 + 5 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$u = x^2 + 5 \Rightarrow x^2 = u - 5$

$\Rightarrow 2x^2 - 1 = 2(u-5) - 1 \Rightarrow 2x^2 - 1 = 2u - 11$

$\int \frac{2x^3 - x}{\sqrt{x^2+5}}\,\mathrm{d}x = \int \frac{2x^3 - x}{\sqrt{u}} \times \frac{1}{2x}\,\mathrm{d}u = \int \frac{2x^2 - 1}{2u^{\frac{1}{2}}}\,\mathrm{d}u$

$= \int \frac{2u - 11}{2u^{\frac{1}{2}}}\,\mathrm{d}u = \int \left(\frac{2u}{2u^{\frac{1}{2}}} - \frac{11}{2u^{\frac{1}{2}}}\right)\mathrm{d}u$

$= \int \left(u^{\frac{1}{2}} - \frac{11}{2}u^{-\frac{1}{2}}\right)\mathrm{d}u$

$= \frac{2}{3}u^{\frac{3}{2}} - 11u^{\frac{1}{2}} + c$

$= \frac{2}{3}(x^2+5)^{\frac{3}{2}} - 11(x^2+5)^{\frac{1}{2}} + c$

19)

$\int_0^1 \frac{x^7}{4-x^4}\,\mathrm{d}x$

$u = 4 - x^4 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -4x^3 \Rightarrow \mathrm{d}x = -\frac{1}{4x^3}\mathrm{d}u$

$u = 4 - x^4 \Rightarrow x^4 = 4 - u$

Change limits:

$x = 1 \Rightarrow u = 4 - 1^4 = 3$

$x = 0 \Rightarrow u = 4 - 0^4 = 4$

$\int_0^1 \frac{x^7}{4-x^4}\,\mathrm{d}x = \int_4^3 \frac{x^7}{u} \times -\frac{1}{4x^3}\,\mathrm{d}u = \int_3^4 \frac{x^4}{4u}\,\mathrm{d}u$

$= \int_3^4 \frac{4-u}{4u}\,\mathrm{d}u = \int_3^4 \left(\frac{4}{4u} - \frac{u}{4u}\right)\mathrm{d}u = \int_3^4 \left(\frac{1}{u} - \frac{1}{4}\right)\mathrm{d}u$

$= \left[\ln|u| - \frac{1}{4}u\right]_3^4$

$= \left(\ln 4 - \frac{1}{4} \times 4\right) - \left(\ln 3 - \frac{1}{4} \times 3\right)$

$= (\ln 4 - 1) - \left(\ln 3 - \frac{3}{4}\right) = \ln 4 - \ln 3 - \frac{1}{4} = \ln\frac{4}{3} - \frac{1}{4}$

20)

$\int_0^1 \frac{20x^9}{(x^5+3)^2}\,\mathrm{d}x$

$u = x^5 + 3 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 5x^4 \Rightarrow \mathrm{d}x = \frac{1}{5x^4}\mathrm{d}u$

$u = x^5 + 3 \Rightarrow x^5 = u - 3 \Rightarrow 20x^5 = 20u - 60$

Change limits:

$x = 1 \Rightarrow u = 1^5 + 3 = 4$

$x = 0 \Rightarrow u = 0^5 + 3 = 3$

$\int_0^1 \frac{20x^9}{(x^5+3)^2}\,\mathrm{d}x = \int_0^1 \frac{20x^9}{u^2} \times \frac{1}{5x^4}\,\mathrm{d}u = \int_3^4 \frac{20x^5}{5u^2}\,\mathrm{d}u$

$= \int_3^4 \frac{20u - 60}{5u^2}\,\mathrm{d}u = \int_3^4 \left(\frac{20u}{5u^2} - \frac{60}{5u^2}\right)\mathrm{d}u$

$= \int_3^4 \left(\frac{4}{u} - 12u^{-2}\right)\mathrm{d}u$

$= [4\ln|u| + 12u^{-1}]_3^4$

$= (4\ln 4 + 12 \times 4^{-1}) - (4\ln 3 + 12 \times 3^{-1})$

$= (4\ln 4 + 3) - (4\ln 3 + 4) = 4\ln 4 - 4\ln 3 - 1$

$= 4(\ln 4 - \ln 3) - 1 = 4\ln\frac{4}{3} - 1 = \ln\left(\frac{4}{3}\right)^4 - 1$

$= \ln\frac{256}{81} - 1$

21)

$\int \frac{2}{2\sqrt{x}-1}\,\mathrm{d}x$

$u = 2x^{\frac{1}{2}} - 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = x^{-\frac{1}{2}} \Rightarrow \mathrm{d}x = x^{\frac{1}{2}}\mathrm{d}u$

$u = 2x^{\frac{1}{2}} - 1 \Rightarrow x^{\frac{1}{2}} = \frac{u+1}{2} \Rightarrow x^{-\frac{1}{2}} = \frac{2}{u+1}$

$\therefore \mathrm{d}x = x^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = \left(\frac{u+1}{2}\right)\mathrm{d}u$

$\int \frac{2}{2\sqrt{x}-1}\,\mathrm{d}x = \int \frac{2}{u} \times \left(\frac{u+1}{2}\right)\mathrm{d}u = \int \frac{u+1}{u}\,\mathrm{d}u$

$= \int \left(\frac{u}{u} + \frac{1}{u}\right)\mathrm{d}u = \int \left(1 + \frac{1}{u}\right)\mathrm{d}u$

$= u + \ln|u| + c$

$= (2\sqrt{x} - 1) + \ln|2\sqrt{x} - 1| + c$

22)

$\int_1^4 \frac{2}{\sqrt{x}+1}\,\mathrm{d}x$

$u = x^{\frac{1}{2}} + 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}x^{-\frac{1}{2}} \Rightarrow \mathrm{d}x = 2x^{\frac{1}{2}}\mathrm{d}u$

$u = x^{\frac{1}{2}} + 1 \Rightarrow x^{\frac{1}{2}} = u - 1$

$\therefore \mathrm{d}x = 2x^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = 2(u-1)\,\mathrm{d}u$

Change limits:

$x = 4 \Rightarrow u = \sqrt{4} + 1 = 3$

$x = 1 \Rightarrow u = \sqrt{1} + 1 = 2$

$\int_1^4 \frac{2}{\sqrt{x}+1}\,\mathrm{d}x = \int_2^3 \frac{2}{u} \times 2(u-1)\,\mathrm{d}u = \int_2^3 \frac{4(u-1)}{u}\,\mathrm{d}u$

$= \int_2^3 \frac{4u-4}{u}\,\mathrm{d}u = \int_2^3 \left(\frac{4u}{u} - \frac{4}{u}\right)\mathrm{d}u = \int_2^3 \left(4 - \frac{4}{u}\right)\mathrm{d}u$

$= [4u - 4\ln|u|]_2^3$

$= (4 \times 3 - 4\ln 3) - (4 \times 2 - 4\ln 2)$

$= (12 - 4\ln 3) - (8 - 4\ln 2) = 4 + 4\ln 2 - 4\ln 3$

$= 4 + 4(\ln 2 - \ln 3) = 4 + 4\ln\frac{2}{3}$

23)

$\int_2^5 \frac{2x^2}{\sqrt{x-1}}\,\mathrm{d}x$

$u = (x-1)^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}(x-1)^{-\frac{1}{2}}$

$\Rightarrow \mathrm{d}x = 2(x-1)^{\frac{1}{2}}\mathrm{d}u$

$u = (x-1)^{\frac{1}{2}} \Rightarrow x = u^2 + 1$

$\therefore \mathrm{d}x = 2(x-1)^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

Change limits:

$x = 5 \Rightarrow u = \sqrt{5-1} = 2$

$x = 2 \Rightarrow u = \sqrt{2-1} = 1$

$\int_2^5 \frac{2x^2}{\sqrt{x-1}}\,\mathrm{d}x = \int_1^2 \frac{2(u^2+1)^2}{u} \times 2u\,\mathrm{d}u = \int_1^2 4(u^2+1)^2\,\mathrm{d}u$

$= \int_1^2 4(u^4 + 2u^2 + 1)\,\mathrm{d}u = \int_1^2 (4u^4 + 8u^2 + 4)\,\mathrm{d}u$

$= \left[\frac{4}{5}u^5 + \frac{8}{3}u^3 + 4u\right]_1^2$

$= \left(\frac{4}{5} \times 2^5 + \frac{8}{3} \times 2^3 + 4 \times 2\right) - \left(\frac{4}{5} \times 1^5 + \frac{8}{3} \times 1^3 + 4 \times 1\right)$

$= \left(\frac{824}{15}\right) - \left(\frac{112}{15}\right) = \frac{712}{15}$

24)

$\int \frac{3x}{1+\sqrt{x}}\,\mathrm{d}x$

$u = 1 + x^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}x^{-\frac{1}{2}} \Rightarrow \mathrm{d}x = 2x^{\frac{1}{2}}\mathrm{d}u$

$u = 1 + x^{\frac{1}{2}} \Rightarrow x^{\frac{1}{2}} = u - 1 \Rightarrow x = (u-1)^2$

$\therefore \mathrm{d}x = 2x^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = 2(u-1)\,\mathrm{d}u$

$\int \frac{3x}{1+\sqrt{x}}\,\mathrm{d}x = \int \frac{3(u-1)^2}{u} \times 2(u-1)\,\mathrm{d}u = \int \frac{6(u-1)^3}{u}\,\mathrm{d}u$

$= \int \frac{6(u^3-3u^2+3u-1)}{u}\,\mathrm{d}u = \int \left(\frac{6u^3}{u} - \frac{18u^2}{u} + \frac{18u}{u} - \frac{6}{u}\right)\,\mathrm{d}u$

$= \int \left(6u^2 - 18u + 18 - \frac{6}{u}\right)\,\mathrm{d}u$

$= 2u^3 - 9u^2 + 18u - 6\ln|u| + c$

$= 2(1+\sqrt{x})^3 - 9(1+\sqrt{x})^2 + 18(1+\sqrt{x}) - 6\ln|1+\sqrt{x}| + c$

25)

$\int_1^4 \frac{1}{\sqrt{x}(2+\sqrt{x})}\,\mathrm{d}x$

$u = x^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}x^{-\frac{1}{2}} \Rightarrow \mathrm{d}x = 2x^{\frac{1}{2}}\mathrm{d}u$

$u = x^{\frac{1}{2}} \Rightarrow x = u^2$

$\therefore \mathrm{d}x = 2x^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

Change limits:

$x = 4 \Rightarrow u = \sqrt{4} = 2$

$x = 1 \Rightarrow u = \sqrt{1} = 1$

$\int_1^4 \frac{1}{\sqrt{x}(2+\sqrt{x})}\,\mathrm{d}x = \int_1^2 \frac{1}{u(2+u)} \times 2u\,\mathrm{d}u = \int_1^2 \frac{2}{2+u}\,\mathrm{d}u$

$= [2\ln|2+u|]_1^2$

$= (2\ln|2+2|) - (2\ln|2+1|) = (2\ln 4) - (2\ln 3)$

$= 2(\ln 4 - \ln 3) = 2\ln\frac{4}{3}$

26)

$\int_1^9 \frac{1+\sqrt{x}}{1+2\sqrt{x}}\,\mathrm{d}x$

$u = 1 + 2x^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = x^{-\frac{1}{2}} \Rightarrow \mathrm{d}x = x^{\frac{1}{2}}\mathrm{d}u$

$u = 1 + 2x^{\frac{1}{2}} \Rightarrow x^{\frac{1}{2}} = \frac{u-1}{2}$

$\therefore \mathrm{d}x = x^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = \left(\frac{u-1}{2}\right)\mathrm{d}u$

Change limits:

$x = 9 \Rightarrow u = 1 + 2\sqrt{9} = 7$

$x = 1 \Rightarrow u = 1 + 2\sqrt{1} = 3$

$\int_1^9 \frac{1+\sqrt{x}}{1+2\sqrt{x}}\,\mathrm{d}x = \int_3^7 \frac{1+\frac{u-1}{2}}{u} \times \left(\frac{u-1}{2}\right)\,\mathrm{d}u$

$= \int_3^7 \frac{2+(u-1)}{2u} \times \left(\frac{u-1}{2}\right) \mathrm{d}u = \int_3^7 \frac{2(u-1)+(u-1)^2}{4u} \mathrm{d}u$

$= \int_3^7 \frac{2u-2+(u^2-2u+1)}{4u} \mathrm{d}u = \int_3^7 \frac{u^2-1}{4u} \mathrm{d}u$

$= \int_3^7 \left(\frac{u^2}{4u} - \frac{1}{4u}\right) \mathrm{d}u = \int_3^7 \left(\frac{1}{4}u - \frac{1}{4u}\right) \mathrm{d}u$

$= \left[\frac{1}{8}u^2 - \frac{1}{4}\ln|u|\right]_3^7$

$= \left(\frac{1}{8} \times 7^2 - \frac{1}{4}\ln 7\right) - \left(\frac{1}{8} \times 3^2 - \frac{1}{4}\ln 3\right)$

$= \left(\frac{49}{8} - \frac{1}{4}\ln 7\right) - \left(\frac{9}{8} - \frac{1}{4}\ln 3\right) = 5 + \frac{1}{4}(\ln 3 - \ln 7)$

$= 5 + \frac{1}{4}\ln\frac{3}{7} = 5 - \frac{1}{4}\ln\frac{7}{3}$

27)

$\int_0^3 \frac{8-x}{3+\sqrt{x+1}} \mathrm{d}x$

$u = (x+1)^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}(x+1)^{-\frac{1}{2}}$

$\Rightarrow \mathrm{d}x = 2(x+1)^{\frac{1}{2}}\mathrm{d}u$

$u = (x+1)^{\frac{1}{2}} \Rightarrow x = u^2 - 1$

$\Rightarrow 8 - x = 8 - (u^2 - 1) \Rightarrow 8 - x = 9 - u^2$

$\therefore \mathrm{d}x = 2(x+1)^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

Change limits:

$x = 3 \Rightarrow u = \sqrt{3+1} = 2$

$x = 0 \Rightarrow u = \sqrt{0+1} = 1$

$$\int_0^3 \frac{8-x}{3+\sqrt{x+1}} \mathrm{d}x = \int_1^2 \frac{9-u^2}{3+u} \times 2u\,\mathrm{d}u$$
$$= \int_1^2 \frac{(3-u)(3+u)}{3+u} \times 2u\,\mathrm{d}u$$
$$= \int_1^2 (3-u) \times 2u\,\mathrm{d}u = \int_1^2 (6u - 2u^2)\,\mathrm{d}u$$

$= \left[3u^2 - \frac{2}{3}u^3\right]_1^2$

$= \left(3 \times 2^2 - \frac{2}{3} \times 2^3\right) - \left(3 \times 1^2 - \frac{2}{3} \times 1^3\right)$

$= \left(\frac{20}{3}\right) - \left(\frac{7}{3}\right) = \frac{13}{3}$

28 a)

$\int_0^{16} \frac{4}{1+\sqrt{x}} \mathrm{d}x$

$u^2 = x \Rightarrow u = x^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}x^{-\frac{1}{2}} \Rightarrow \mathrm{d}x = 2x^{\frac{1}{2}}\mathrm{d}u$

$\Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

Change limits:

$x = 16 \Rightarrow u = \sqrt{16} = 4$

$x = 0 \Rightarrow u = \sqrt{0} = 0$

$\int_0^{16} \frac{4}{1+\sqrt{x}} \mathrm{d}x = \int_0^4 \frac{4}{1+u} \times 2u\,\mathrm{d}u = \int_0^4 \frac{8u}{1+u} \mathrm{d}u$

$= \int_0^4 \left(\frac{8u+8}{1+u} - \frac{8}{1+u}\right) \mathrm{d}u = \int_0^4 \left(\frac{8(1+u)}{1+u} - \frac{8}{1+u}\right) \mathrm{d}u$

$= \int_0^4 \left(8 - \frac{8}{1+u}\right) \mathrm{d}u$

$= [8u - 8\ln|1+u|]_0^4$

$= (8 \times 4 - 8\ln|1+4|) - (8 \times 0 - 8\ln|1+0|)$

$= (32 - 8\ln 5) - (0 - 0) = 32 - 8\ln 5$

28 b)

$\int_1^3 \frac{8x}{\sqrt{4x-3}} \mathrm{d}x$

$u^2 = 4x - 3 \Rightarrow u = (4x-3)^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2(4x-3)^{-\frac{1}{2}}$

$\Rightarrow \mathrm{d}x = \frac{1}{2}(4x-3)^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = \frac{1}{2}u\,\mathrm{d}u$

$u^2 = 4x - 3 \Rightarrow 4x = u^2 + 3 \Rightarrow 8x = 2u^2 + 6$

Change limits:

$x = 3 \Rightarrow u = \sqrt{4 \times 3 - 3} = 3$

$x = 1 \Rightarrow u = \sqrt{4 \times 1 - 3} = 1$

$\int_1^3 \frac{8x}{\sqrt{4x-3}} \mathrm{d}x = \int_1^3 \frac{2u^2+6}{u} \times \frac{1}{2}u\,\mathrm{d}u = \int_1^3 \frac{1}{2}(2u^2+6)\,\mathrm{d}u$

$= \int_1^3 (u^2+3)\,\mathrm{d}u$

$= \left[\frac{1}{3}u^3 + 3u\right]_1^3$

$= \left(\frac{1}{3} \times 3^3 + 3 \times 3\right) - \left(\frac{1}{3} \times 1^3 + 3 \times 1\right)$

$= (18) - \left(\frac{10}{3}\right) = \frac{44}{3}$

29 a)

$\int_1^4 \frac{12}{4\sqrt{x}+3x} \mathrm{d}x$

$u = x^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}x^{-\frac{1}{2}} \Rightarrow \mathrm{d}x = 2x^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

$u = x^{\frac{1}{2}} \Rightarrow x = u^2$

Change limits:

$x = 4 \Rightarrow u = \sqrt{4} = 2$

$x = 1 \Rightarrow u = \sqrt{1} = 1$

$\int_1^4 \frac{12}{4\sqrt{x}+3x} \mathrm{d}x = \int_1^2 \frac{12}{4u+3u^2} \times 2u\,\mathrm{d}u = \int_1^2 \frac{24u}{4u+3u^2}\,\mathrm{d}u$

$= \int_1^2 \frac{24}{4+3u}\,\mathrm{d}u$

$= [8\ln|4+3u|]_1^2$

$= (8\ln|4+3\times 2|) - (8\ln|4+3\times 1|)$

$= (8\ln 10) - (8\ln 7) = 8(\ln 10 - \ln 7) = 8\ln\frac{10}{7}$

29 b)

$\int_1^4 \frac{12}{4\sqrt{x}+3x} \mathrm{d}x$

$u = 3x^{\frac{1}{2}} + 4 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{3}{2}x^{-\frac{1}{2}} \Rightarrow \mathrm{d}x = \frac{2}{3}x^{\frac{1}{2}}\mathrm{d}u$

Change limits:

$x = 4 \Rightarrow u = 3\sqrt{4} + 4 = 10$

$x = 1 \Rightarrow u = 3\sqrt{1} + 4 = 7$

$$\begin{aligned}\int_1^4 \frac{12}{4\sqrt{x} + 3x}\,\mathrm{d}x &= \int_1^4 \frac{12}{\sqrt{x}(3\sqrt{x} + 4)}\,\mathrm{d}x \\ &= \int_7^{10} \frac{12}{x^{\frac{1}{2}}u} \times \frac{2}{3}x^{\frac{1}{2}}\,\mathrm{d}u \\ &= \int_7^{10} \frac{12}{u} \times \frac{2}{3}\,\mathrm{d}u = \int_7^{10} \frac{8}{u}\,\mathrm{d}u\end{aligned}$$

$= [8\ln|u|]_7^{10}$

$= (8\ln|10|) - (8\ln|7|) = (8\ln 10) - (8\ln 7)$

$= 8(\ln 10 - \ln 7) = 8\ln\frac{10}{7}$

30)

$\lim_{\delta x \to 0} \sum_{x=0}^{2} \frac{x^3}{\sqrt{2x^2+1}}\delta x = \int_0^2 \frac{x^3}{\sqrt{2x^2+1}}\,\mathrm{d}x$

$u^2 = 2x^2 + 1 \Rightarrow u = (2x^2+1)^{\frac{1}{2}}$

$\Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x(2x^2+1)^{-\frac{1}{2}} \Rightarrow \mathrm{d}x = \frac{(2x^2+1)^{\frac{1}{2}}}{2x}\mathrm{d}u$

$\Rightarrow \mathrm{d}x = \frac{u}{2x}\mathrm{d}u$

$u^2 = 2x^2 + 1 \Rightarrow x^2 = \frac{u^2-1}{2}$

Change limits:

$x = 2 \Rightarrow u = \sqrt{2 \times 2^2 + 1} = 3$

$x = 0 \Rightarrow u = \sqrt{2 \times 0^2 + 1} = 1$

$$\begin{aligned}\int_0^2 \frac{x^3}{\sqrt{2x^2+1}}\,\mathrm{d}x = \int_1^3 \frac{x^3}{u} \times \frac{u}{2x}\,\mathrm{d}u &= \int_1^3 \frac{1}{2}x^2\,\mathrm{d}u \\ &= \int_1^3 \frac{1}{2}\left(\frac{u^2-1}{2}\right)\,\mathrm{d}u = \int_1^3 \frac{1}{4}(u^2-1)\,\mathrm{d}u \\ &= \int_1^3 \left(\frac{1}{4}u^2 - \frac{1}{4}\right)\,\mathrm{d}u\end{aligned}$$

$= \left[\frac{1}{12}u^3 - \frac{1}{4}u\right]_1^3$

$= \left(\frac{1}{12} \times 3^3 - \frac{1}{4} \times 3\right) - \left(\frac{1}{12} \times 1^3 - \frac{1}{4} \times 1\right)$

$= \left(\frac{3}{2}\right) - \left(-\frac{1}{6}\right) = \frac{5}{3}$

31 a)

$\int \cos x \sin^4 x\,\mathrm{d}x$

$u = \sin x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \cos x \Rightarrow \mathrm{d}x = \frac{1}{\cos x}\mathrm{d}u$

$\int \cos x \sin^4 x\,\mathrm{d}x = \int \cos x \times u^4 \times \frac{1}{\cos x}\,\mathrm{d}u = \int u^4\,\mathrm{d}u$

$= \frac{1}{5}u^5 + c = \frac{1}{5}\sin^5 x + c$

31 b)

$\int 8x\mathrm{e}^{x^2+1}\,\mathrm{d}x$

$u = x^2 + 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$\int 8xe^{x^2+1}\,dx = \int 8x \times e^u \times \frac{1}{2x}\,du = \int 4e^u\,du$

$= 4e^u + c = 4e^{x^2+1} + c$

31 c)

$\int \sin x\, e^{\cos x}\,dx$

$u = \cos x \Rightarrow \frac{du}{dx} = -\sin x \Rightarrow dx = -\frac{1}{\sin x}du$

$\int \sin x\, e^{\cos x}\,dx = \int \sin x \times e^u \times -\frac{1}{\sin x}\,du$

$= \int -e^u\,du$

$= -e^u + c = -e^{\cos x} + c$

31 d)

$\int \frac{\sec^2 x}{\tan x}\,dx$

$u = \tan x \Rightarrow \frac{du}{dx} = \sec^2 x \Rightarrow dx = \frac{1}{\sec^2 x}du$

$\int \frac{\sec^2 x}{\tan x}\,dx = \int \frac{\sec^2 x}{u} \times \frac{1}{\sec^2 x}\,du = \int \frac{1}{u}\,du$

$= \ln|u| + c = \ln|\tan x| + c$

32)

$\int_0^{\frac{\pi}{2}} \frac{6\cos x}{9+\sin x}\,dx$

$u = 9 + \sin x \Rightarrow \frac{du}{dx} = \cos x \Rightarrow dx = \frac{1}{\cos x}du$

Change limits:

$x = \frac{\pi}{2} \Rightarrow u = 9 + \sin\frac{\pi}{2} = 10$

$x = 0 \Rightarrow u = 9 + \sin 0 = 9$

$\int_0^{\frac{\pi}{2}} \frac{6\cos x}{9+\sin x}\,dx = \int_9^{10} \frac{6\cos x}{u} \times \frac{1}{\cos x}\,du = \int_9^{10} \frac{6}{u}\,du$

$= [6\ln|u|]_9^{10}$

$= (6\ln|10|) - (6\ln|9|) = (6\ln 10) - (6\ln 9)$

$= 6(\ln 10 - \ln 9) = 6\ln\frac{10}{9}$

33)

$\int \frac{5e^x}{5+e^x}\,dx$

$u = 5 + e^x \Rightarrow \frac{du}{dx} = e^x \Rightarrow dx = \frac{1}{e^x}du$

$\int \frac{5e^x}{5+e^x}\,dx = \int \frac{5e^x}{u} \times \frac{1}{e^x}\,du = \int \frac{5}{u}\,du$

$= 5\ln|u| + c$

$= 5\ln|5 + e^x| + c$

34 a)

$\int_0^{\ln 2} \frac{3e^{2x}}{3+e^x}\,dx$

$u = 3 + e^x \Rightarrow \frac{du}{dx} = e^x \Rightarrow dx = \frac{1}{e^x}du$

$u = 3 + e^x \Rightarrow e^x = u - 3$

Change limits:

$x = \ln 2 \Rightarrow u = 3 + e^{\ln 2} = 3 + 2 = 5$

$x = 0 \Rightarrow u = 3 + \mathrm{e}^0 = 3 + 1 = 4$

$\int_0^{\ln 2} \frac{3\mathrm{e}^{2x}}{3+\mathrm{e}^x}\,\mathrm{d}x = \int_4^5 \frac{3\mathrm{e}^{2x}}{u} \times \frac{1}{\mathrm{e}^x}\,\mathrm{d}u = \int_4^5 \frac{3\mathrm{e}^x}{u}\,\mathrm{d}u$

$= \int_4^5 \frac{3(u-3)}{u}\,\mathrm{d}u = \int_4^5 \frac{3u-9}{u}\,\mathrm{d}u = \int_4^5 \left(\frac{3u}{u} - \frac{9}{u}\right)\,\mathrm{d}u$

$= \int_4^5 \left(3 - \frac{9}{u}\right)\,\mathrm{d}u$

$= [3u - 9\ln|u|]_4^5$

$= (3 \times 5 - 9\ln|5|) - (3 \times 4 - 9\ln|4|)$

$= (15 - 9\ln 5) - (12 - 9\ln 4)$

$= 3 + 9\ln 4 - 9\ln 5 = 3 + 9(\ln 4 - \ln 5)$

$= 3 + 9\ln\frac{4}{5} = 3 - 9\ln\frac{5}{4}$

34 b)

$\int_{\ln 2}^{\ln 3} \frac{8\mathrm{e}^x}{(2+\mathrm{e}^x)^2}\,\mathrm{d}x$

$u = 2 + \mathrm{e}^x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x \Rightarrow \mathrm{d}x = \frac{1}{\mathrm{e}^x}\mathrm{d}u$

$u = 2 + \mathrm{e}^x \Rightarrow \mathrm{e}^x = u - 2$

Change limits:

$x = \ln 3 \Rightarrow u = 2 + \mathrm{e}^{\ln 3} = 2 + 3 = 5$

$x = \ln 2 \Rightarrow u = 2 + \mathrm{e}^{\ln 2} = 2 + 2 = 4$

$\int_{\ln 2}^{\ln 3} \frac{8\mathrm{e}^x}{(2+\mathrm{e}^x)^2}\,\mathrm{d}x = \int_4^5 \frac{8\mathrm{e}^x}{u^2} \times \frac{1}{\mathrm{e}^x}\,\mathrm{d}u = \int_4^5 \frac{8}{u^2}\,\mathrm{d}u$

$= \int_4^5 8u^{-2}\,\mathrm{d}u$

$= [-8u^{-1}]_4^5 = \left[-\frac{8}{u}\right]_4^5$

$= \left(-\frac{8}{5}\right) - \left(-\frac{8}{4}\right) = \left(-\frac{8}{5}\right) - (-2) = \frac{2}{5}$

35 a)

$\int_0^{\frac{\pi}{4}} 8\cos 2x \sin^5 2x\,\mathrm{d}x$

$u = \sin 2x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2\cos 2x \Rightarrow \mathrm{d}x = \frac{1}{2\cos 2x}\mathrm{d}u$

Change limits:

$x = \frac{\pi}{4} \Rightarrow u = \sin\left(2 \times \frac{\pi}{4}\right) = 1$

$x = 0 \Rightarrow u = \sin(2 \times 0) = 0$

$\int_0^{\frac{\pi}{4}} 8\cos 2x \sin^5 2x\,\mathrm{d}x = \int_0^1 8\cos 2x \times u^5 \times \frac{1}{2\cos 2x}\,\mathrm{d}u = \int_0^1 4u^5\,\mathrm{d}u$

$= \left[\frac{2}{3}u^6\right]_0^1$

$= \left(\frac{2}{3} \times 1^6\right) - \left(\frac{2}{3} \times 0^6\right) = \left(\frac{2}{3}\right) - (0) = \frac{2}{3}$

35 b)

$\int_0^{\frac{\pi}{8}} \sin^2 2x \cos^3 2x\,\mathrm{d}x$

$u = \sin 2x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2\cos 2x \Rightarrow \mathrm{d}x = \frac{1}{2\cos 2x}\mathrm{d}u$

Change limits:

$x = \frac{\pi}{8} \Rightarrow u = \sin\left(2 \times \frac{\pi}{8}\right) = \frac{\sqrt{2}}{2}$

$x = 0 \Rightarrow u = \sin(2 \times 0) = 0$

$\int_0^{\frac{\pi}{8}} \sin^2 2x \cos^3 2x \, \mathrm{d}x = \int_0^{\frac{\sqrt{2}}{2}} u^2 \times \cos^3 2x \times \frac{1}{2\cos 2x} \, \mathrm{d}u$

$= \int_0^{\frac{\sqrt{2}}{2}} \frac{1}{2} u^2 \cos^2 2x \, \mathrm{d}u = \int_0^{\frac{\sqrt{2}}{2}} \frac{1}{2} u^2 (1 - \sin^2 2x) \, \mathrm{d}u$

$= \int_0^{\frac{\sqrt{2}}{2}} \frac{1}{2} u^2 (1 - u^2) \, \mathrm{d}u = \int_0^{\frac{\sqrt{2}}{2}} \left(\frac{1}{2} u^2 - \frac{1}{2} u^4 \right) \mathrm{d}u$

$= \left[\frac{1}{6} u^3 - \frac{1}{10} u^5 \right]_0^{\frac{\sqrt{2}}{2}}$

$= \left(\frac{1}{6} \times \left(\frac{\sqrt{2}}{2} \right)^3 - \frac{1}{10} \times \left(\frac{\sqrt{2}}{2} \right)^5 \right) - \left(\frac{1}{6} \times 0^5 - \frac{1}{10} \times 0^5 \right)$

$= \left(\frac{\sqrt{2}}{24} - \frac{\sqrt{2}}{80} \right) - (0 - 0) = \frac{7\sqrt{2}}{240}$

36 a)

$\int_0^{\frac{\pi}{4}} \sec^2 x \, \mathrm{e}^{\tan x + 1} \, \mathrm{d}x$

$u = \tan x + 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \sec^2 x \Rightarrow \mathrm{d}x = \frac{1}{\sec^2 x} \mathrm{d}u$

Change limits:

$x = \frac{\pi}{4} \Rightarrow u = \tan \frac{\pi}{4} + 1 = 2$

$x = 0 \Rightarrow u = \tan 0 + 1 = 1$

$\int_0^{\frac{\pi}{4}} \sec^2 x \, \mathrm{e}^{\tan x + 1} \, \mathrm{d}x = \int_1^2 \sec^2 x \times \mathrm{e}^u \times \frac{1}{\sec^2 x} \, \mathrm{d}u$

$= \int_1^2 \mathrm{e}^u \, \mathrm{d}u$

$= [\mathrm{e}^u]_1^2$

$= (\mathrm{e}^2) - (\mathrm{e}^1) = \mathrm{e}^2 - \mathrm{e}$

36 b)

$\int_0^{\frac{\pi}{4}} \sec^4 x \sqrt[3]{\tan x} \, \mathrm{d}x$

$u = \tan x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \sec^2 x \Rightarrow \mathrm{d}x = \frac{1}{\sec^2 x} \mathrm{d}u$

Change limits:

$x = \frac{\pi}{4} \Rightarrow u = \tan \frac{\pi}{4} = 1$

$x = 0 \Rightarrow u = \tan 0 = 0$

$\int_0^{\frac{\pi}{4}} \sec^4 x (\tan x)^{\frac{1}{3}} \, \mathrm{d}x = \int_0^1 \sec^4 x \times u^{\frac{1}{3}} \times \frac{1}{\sec^2 x} \, \mathrm{d}u$

$= \int_0^1 \sec^2 x \times u^{\frac{1}{3}} \, \mathrm{d}u = \int_0^1 (\tan^2 x + 1) \times u^{\frac{1}{3}} \, \mathrm{d}u$

$= \int_0^1 (u^2 + 1) \times u^{\frac{1}{3}} \, \mathrm{d}u = \int_0^1 (u^{\frac{7}{3}} + u^{\frac{1}{3}}) \, \mathrm{d}u$

$= \left[\frac{3}{10} u^{\frac{10}{3}} + \frac{3}{4} u^{\frac{4}{3}} \right]_0^1$

$= \left(\frac{3}{10} \times 1^{\frac{10}{3}} + \frac{3}{4} \times 1^{\frac{4}{3}} \right) - \left(\frac{3}{10} \times 0^{\frac{10}{3}} + \frac{3}{4} \times 0^{\frac{4}{3}} \right)$

$= \left(\frac{21}{20} \right) - (0) = \frac{21}{20}$

37)

$\int \frac{1}{\cos^2 x (1 + \tan x)^2} \, \mathrm{d}x$

$u = 1 + \tan x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \sec^2 x \Rightarrow \mathrm{d}x = \frac{1}{\sec^2 x} \mathrm{d}u \Rightarrow \mathrm{d}x = \cos^2 x \, \mathrm{d}u$

$\int \frac{1}{\cos^2 x (1 + \tan x)^2} \, \mathrm{d}x = \int \frac{1}{\cos^2 x \times u^2} \times \cos^2 x \, \mathrm{d}u$

$= \int \frac{1}{u^2}\,\mathrm{d}u = \int u^{-2}\,\mathrm{d}u$

$= -u^{-1} + c = -\frac{1}{u} + c = -\frac{1}{1+\tan x} + c$

38 a)

$\int_1^2 \frac{5^x}{(5^x-1)^3}\,\mathrm{d}x$

$u = 5^x - 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = (\ln 5)5^x \Rightarrow \mathrm{d}x = \frac{1}{(\ln 5)5^x}\mathrm{d}u$

Change limits:

$x = 2 \Rightarrow u = 5^2 - 1 = 24$

$x = 1 \Rightarrow u = 5^1 - 1 = 4$

$\int_1^2 \frac{5^x}{(5^x-1)^3}\,\mathrm{d}x = \int_4^{24} \frac{5^x}{u^3} \times \frac{1}{(\ln 5)5^x}\,\mathrm{d}u = \int_4^{24} \frac{1}{u^3} \times \frac{1}{\ln 5}\,\mathrm{d}u$

$= \frac{1}{\ln 5}\int_4^{24} u^{-3}\,\mathrm{d}u$

$= \frac{1}{\ln 5}\left[-\frac{1}{2}u^{-2}\right]_4^{24} = \frac{1}{\ln 5}\left[-\frac{1}{2u^2}\right]_4^{24}$

$= \frac{1}{\ln 5}\left(\left(-\frac{1}{2\times 24^2}\right) - \left(-\frac{1}{2\times 4^2}\right)\right)$

$= \frac{1}{\ln 5}\left(\left(-\frac{1}{1152}\right) - \left(-\frac{1}{32}\right)\right)$

$= \frac{1}{\ln 5}\left(\left(-\frac{1}{1152}\right) - \left(-\frac{1}{32}\right)\right) = \frac{1}{\ln 5} \times \frac{35}{1152} = \frac{35}{1152\ln 5}$

38 b)

$\int_1^2 \frac{5^x}{(5^x-1)^3}\,\mathrm{d}x$

$u = 5^x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = (\ln 5)5^x \Rightarrow \mathrm{d}x = \frac{1}{(\ln 5)5^x}\mathrm{d}u$

Change limits:

$x = 2 \Rightarrow u = 5^2 = 25$

$x = 1 \Rightarrow u = 5^1 = 5$

$$\int_1^2 \frac{5^x}{(5^x-1)^3}\,\mathrm{d}x = \int_5^{25} \frac{5^x}{(u-1)^3} \times \frac{1}{(\ln 5)5^x}\,\mathrm{d}u$$
$$= \int_5^{25} \frac{1}{(u-1)^3} \times \frac{1}{\ln 5}\,\mathrm{d}u = \frac{1}{\ln 5}\int_5^{25} (u-1)^{-3}\,\mathrm{d}u$$

$= \frac{1}{\ln 5}\left[-\frac{1}{2}(u-1)^{-2}\right]_5^{25} = \frac{1}{\ln 5}\left[-\frac{1}{2(u-1)^2}\right]_5^{25}$

$= \frac{1}{\ln 5}\left(\left(-\frac{1}{2\times(25-1)^2}\right) - \left(-\frac{1}{2\times(5-1)^2}\right)\right)$

$= \frac{1}{\ln 5}\left(\left(-\frac{1}{1152}\right) - \left(-\frac{1}{32}\right)\right)$

$= \frac{1}{\ln 5}\left(\left(-\frac{1}{1152}\right) - \left(-\frac{1}{32}\right)\right) = \frac{1}{\ln 5} \times \frac{35}{1152} = \frac{35}{1152\ln 5}$

39 a)

$\int_0^1 8^x(8^x+1)^{\frac{3}{2}}\,\mathrm{d}x$

$u = 8^x + 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = (\ln 8)8^x \Rightarrow \mathrm{d}x = \frac{1}{(\ln 8)8^x}\mathrm{d}u$

Change limits:

$x = 1 \Rightarrow u = 8^1 + 1 = 9$

$x = 0 \Rightarrow u = 8^0 + 1 = 2$

$\int_0^1 8^x(8^x+1)^{\frac{3}{2}}\,\mathrm{d}x = \int_2^9 8^x \times u^{\frac{3}{2}} \times \frac{1}{(\ln 8)8^x}\,\mathrm{d}u$

$= \int_2^9 u^{\frac{3}{2}} \times \frac{1}{\ln 8}\,\mathrm{d}u = \frac{1}{\ln 8}\int_2^9 u^{\frac{3}{2}}\,\mathrm{d}u$

$= \frac{1}{\ln 8}\left[\frac{2}{5}u^{\frac{5}{2}}\right]_2^9$

$= \frac{1}{\ln 8}\left(\left(\frac{2}{5}\times 9^{\frac{5}{2}}\right) - \left(\frac{2}{5}\times 2^{\frac{5}{2}}\right)\right) = \frac{1}{\ln 8}\left(\left(\frac{486}{5}\right) - \left(\frac{8\sqrt{2}}{5}\right)\right)$

$= \frac{486-8\sqrt{2}}{5\ln 8}$

39 b)

$\int_0^1 8^x(8^x+1)^{\frac{3}{2}}\,\mathrm{d}x$

$u = 8^x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = (\ln 8)8^x \Rightarrow \mathrm{d}x = \frac{1}{(\ln 8)8^x}\mathrm{d}u$

Change limits:

$x = 1 \Rightarrow u = 8^1 = 8$

$x = 0 \Rightarrow u = 8^0 = 1$

$\int_0^1 8^x(8^x+1)^{\frac{3}{2}}\,\mathrm{d}x = \int_1^8 8^x \times (u+1)^{\frac{3}{2}} \times \frac{1}{(\ln 8)8^x}\,\mathrm{d}u$

$= \int_1^8 (u+1)^{\frac{3}{2}} \times \frac{1}{\ln 8}\,\mathrm{d}u = \frac{1}{\ln 8}\int_1^8 (u+1)^{\frac{3}{2}}\,\mathrm{d}u$

$= \frac{1}{\ln 8}\left[\frac{2}{5}(u+1)^{\frac{5}{2}}\right]_1^8$

$= \frac{1}{\ln 8}\left(\left(\frac{2}{5}\times (8+1)^{\frac{5}{2}}\right) - \left(\frac{2}{5}\times (1+1)^{\frac{5}{2}}\right)\right)$

$= \frac{1}{\ln 8}\left(\left(\frac{486}{5}\right) - \left(\frac{8\sqrt{2}}{5}\right)\right) = \frac{486-8\sqrt{2}}{5\ln 8}$

40)

$\int \frac{6}{x(3-\ln x)}\,\mathrm{d}x$

$u = 3 - \ln x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -\frac{1}{x} \Rightarrow \mathrm{d}x = -x\,\mathrm{d}u$

$\int \frac{6}{x(3-\ln x)}\,\mathrm{d}x = \int \frac{6}{x\times u} \times -x\,\mathrm{d}u = \int -\frac{6}{u}\,\mathrm{d}u$

$= -6\ln|u| + c$

$= -6\ln|3-\ln x| + c$

41 a)

$\int_1^{2\mathrm{e}} \frac{4}{x(5+\ln x)^3}\,\mathrm{d}x$

$u = 5 + \ln x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \Rightarrow \mathrm{d}x = x\,\mathrm{d}u$

Change limits:

$x = 2\mathrm{e} \Rightarrow u = 5 + \ln(2\mathrm{e}) = 5 + \ln 2 + \ln \mathrm{e}$

$= 5 + \ln 2 + 1 = 6 + \ln 2$

$x = 1 \Rightarrow u = 5 + \ln 1 = 5 + 0 = 5$

$\int_1^{2\mathrm{e}} \frac{4}{x(5+\ln x)^3}\,\mathrm{d}x = \int_5^{6+\ln 2} \frac{4}{x\times u^3} \times x\,\mathrm{d}u$

$= \int_5^{6+\ln 2} \frac{4}{u^3}\,\mathrm{d}u = \int_5^{6+\ln 2} 4u^{-3}\,\mathrm{d}u$

$= \left[-2u^{-2}\right]_5^{6+\ln 2} = \left[-\frac{2}{u^2}\right]_5^{6+\ln 2}$

$= \left(-\frac{2}{(6+\ln 2)^2}\right) - \left(-\frac{2}{5^2}\right) = \frac{2}{25} - \frac{2}{(6+\ln 2)^2}$

41 b)

$\int_1^{\mathrm{e}} \frac{2\ln x}{x(4-\ln x)^2}\,\mathrm{d}x$

$u = 4 - \ln x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} \Rightarrow \mathrm{d}x = x\,\mathrm{d}u$

$u = 4 - \ln x \Rightarrow \ln x = 4 - u$

Change limits:

$x = \mathrm{e} \Rightarrow u = 4 - \ln \mathrm{e} = 4 - 1 = 3$

$x = 1 \Rightarrow u = 4 - \ln 1 = 4 - 0 = 4$

$\int_1^{\mathrm{e}} \frac{2\ln x}{x(4-\ln x)^2}\,\mathrm{d}x = \int_4^3 \frac{2(4-u)}{x\times u^2} \times x\,\mathrm{d}u = \int_4^3 \frac{2(4-u)}{u^2}\,\mathrm{d}u$

$= \int_3^4 \frac{2(u-4)}{u^2}\,\mathrm{d}u = \int_3^4 \frac{2u-8}{u^2}\,\mathrm{d}u = \int_3^4 \left(\frac{2u}{u^2} - \frac{8}{u^2}\right)\,\mathrm{d}u$

$= \int_3^4 \left(\frac{2}{u} - 8u^{-2}\right)\,\mathrm{d}u$

$= [2\ln|u| + 8u^{-1}]_3^4 = \left[2\ln|u| + \frac{8}{u}\right]_3^4$

$= \left(2\ln|4| + \frac{8}{4}\right) - \left(2\ln|3| + \frac{8}{3}\right)$

$= (2\ln 4 + 2) - \left(2\ln 3 + \frac{8}{3}\right) = 2\ln 4 - 2\ln 3 - \frac{2}{3}$

$= 2(\ln 4 - \ln 3) - \frac{2}{3} = 2\ln\frac{4}{3} - \frac{2}{3} = \ln\frac{16}{9} - \frac{2}{3}$

42 a)

$\int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x$

$x = \sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u\,\mathrm{d}u$

$\int \frac{1}{\sqrt{1-x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{1-\sin^2 u}} \times \cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{\cos^2 u}} \times \cos u\,\mathrm{d}u = \int \frac{1}{\cos u} \times \cos u\,\mathrm{d}u = \int 1\,\mathrm{d}u$

$= u + c$

$= \arcsin x + c$

42 b)

$\int \frac{1}{(1-x^2)^{\frac{3}{2}}}\,\mathrm{d}x$

$x = \sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u\,\mathrm{d}u$

$\int \frac{1}{(1-x^2)^{\frac{3}{2}}}\,\mathrm{d}x = \int \frac{1}{(1-\sin^2 u)^{\frac{3}{2}}} \times \cos u\,\mathrm{d}u$

$= \int \frac{1}{(\cos^2 u)^{\frac{3}{2}}} \times \cos u\,\mathrm{d}u = \int \frac{1}{\cos^3 u} \times \cos u\,\mathrm{d}u$

$= \int \frac{1}{\cos^2 u}\,\mathrm{d}u = \int \sec^2 u\,\mathrm{d}u$

$= \tan u + c = \sqrt{\sec^2 u - 1} + c = \sqrt{\frac{1}{\cos^2 u} - 1} + c$

$= \sqrt{\frac{1}{1-\sin^2 u} - 1} + c = \sqrt{\frac{1}{1-x^2} - 1} + c$

$= \sqrt{\frac{1}{1-x^2} - \frac{1-x^2}{1-x^2}} + c = \sqrt{\frac{x^2}{1-x^2}} + c = \frac{x}{\sqrt{1-x^2}} + c$

43)

$\int_0^{\frac{1}{4}} \frac{1}{\sqrt{1-4x^2}}\,\mathrm{d}x$

$x = \frac{1}{2}\sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{2}\cos u \Rightarrow \mathrm{d}x = \frac{1}{2}\cos u\,\mathrm{d}u$

Change limits:

$x = \frac{1}{4} \Rightarrow \frac{1}{4} = \frac{1}{2}\sin u \Rightarrow \sin u = \frac{1}{2} \Rightarrow u = \arcsin\frac{1}{2} = \frac{\pi}{6}$

$x = 0 \Rightarrow 0 = \frac{1}{2}\sin u \Rightarrow \sin u = 0 \Rightarrow u = 0$

$$\begin{aligned}\int_0^{\frac{1}{4}} \frac{1}{\sqrt{1-4x^2}}\,\mathrm{d}x &= \int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{1-4(\frac{1}{2}\sin u)^2}} \times \frac{1}{2}\cos u\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{1-\sin^2 u}} \times \frac{1}{2}\cos u\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{\cos^2 u}} \times \frac{1}{2}\cos u\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{6}} \frac{1}{\cos u} \times \frac{1}{2}\cos u\,\mathrm{d}u = \int_0^{\frac{\pi}{6}} \frac{1}{2}\,\mathrm{d}u\end{aligned}$$

$= \left[\frac{1}{2}u\right]_0^{\frac{\pi}{6}}$

$= \left(\frac{1}{2} \times \frac{\pi}{6}\right) - \left(\frac{1}{2} \times 0\right) = \left(\frac{\pi}{12}\right) - (0) = \frac{\pi}{12}$

44)

$\int \frac{1}{\sqrt{9-4x^2}}\,\mathrm{d}x$

$x = \frac{3}{2}\sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{3}{2}\cos u \Rightarrow \mathrm{d}x = \frac{3}{2}\cos u\,\mathrm{d}u$

$x = \frac{3}{2}\sin u \Rightarrow \sin u = \frac{2}{3}x \Rightarrow u = \arcsin\left(\frac{2}{3}x\right)$

$\int \frac{1}{\sqrt{9-4x^2}}\,\mathrm{d}x = \int \frac{1}{\sqrt{9-4(\frac{3}{2}\sin u)^2}} \times \frac{3}{2}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{9-9\sin^2 u}} \times \frac{3}{2}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{9(1-\sin^2 u)}} \times \frac{3}{2}\cos u\,\mathrm{d}u$

$= \int \frac{1}{\sqrt{9\cos^2 u}} \times \frac{3}{2}\cos u\,\mathrm{d}u = \int \frac{1}{3\cos u} \times \frac{3}{2}\cos u\,\mathrm{d}u$

$= \int \frac{1}{2}\,\mathrm{d}u = \frac{1}{2}u + c = \frac{1}{2}\arcsin\left(\frac{2}{3}x\right) + c$

45 a)

$\int \frac{1}{1+x^2}\,\mathrm{d}x$

$x = \tan u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u\,\mathrm{d}u$

$\int \frac{1}{1+x^2}\,\mathrm{d}x = \int \frac{1}{1+\tan^2 u} \times \sec^2 u\,\mathrm{d}u$

$= \int \frac{1}{\sec^2 u} \times \sec^2 u\,\mathrm{d}u = \int 1\,\mathrm{d}u$

$= u + c = \arctan x + c$

45 b)

$\int \frac{5}{25+x^2}\,\mathrm{d}x$

$x = 5\tan u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 5\sec^2 u \Rightarrow \mathrm{d}x = 5\sec^2 u\,\mathrm{d}u$

$x = 5\tan u \Rightarrow \tan u = \frac{1}{5}x \Rightarrow u = \arctan\left(\frac{1}{5}x\right)$

$$\int \frac{5}{25+x^2}\,\mathrm{d}x = \int \frac{5}{25+(5\tan u)^2} \times 5\sec^2 u\,\mathrm{d}u$$

$$= \int \frac{5}{25 + 25\tan^2 u} \times 5\sec^2 u \, \mathrm{d}u$$
$$= \int \frac{5}{25(1 + \tan^2 u)} \times 5\sec^2 u \, \mathrm{d}u$$
$$= \int \frac{5}{25\sec^2 u} \times 5\sec^2 u \, \mathrm{d}u = \int 1 \, \mathrm{d}u$$

$= u + c = \arctan\left(\frac{1}{5}x\right) + c$

46)

$\int_0^1 \frac{1}{(1+x^2)^2} \, \mathrm{d}x$

$x = \tan u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u \, \mathrm{d}u$

Change limits:

$x = 1 \Rightarrow 1 = \tan u \Rightarrow u = \arctan 1 = \frac{\pi}{4}$

$x = 0 \Rightarrow 0 = \tan u \Rightarrow u = \arctan 0 = 0$

$$\int_0^1 \frac{1}{(1 + x^2)^2} \, \mathrm{d}x = \int_0^{\frac{\pi}{4}} \frac{1}{(1 + \tan^2 u)^2} \times \sec^2 u \, \mathrm{d}u$$
$$= \int_0^{\frac{\pi}{4}} \frac{1}{(\sec^2 u)^2} \times \sec^2 u \, \mathrm{d}u$$
$$= \int_0^{\frac{\pi}{4}} \frac{1}{\sec^4 u} \times \sec^2 u \, \mathrm{d}u = \int_0^{\frac{\pi}{4}} \frac{1}{\sec^2 u} \, \mathrm{d}u = \int_0^{\frac{\pi}{4}} \cos^2 u \, \mathrm{d}u$$

$\cos 2u = 2\cos^2 u - 1 \Rightarrow \cos^2 u = \frac{1}{2}(\cos 2u + 1)$

$\therefore \int_0^{\frac{\pi}{4}} \cos^2 u \, \mathrm{d}u = \int_0^{\frac{\pi}{4}} \frac{1}{2}(\cos 2u + 1) \, \mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \left(\frac{1}{2}\cos 2u + \frac{1}{2}\right) \, \mathrm{d}u$

$= \left[\frac{1}{4}\sin 2u + \frac{1}{2}u\right]_0^{\frac{\pi}{4}}$

$= \left(\frac{1}{4}\sin\left(2 \times \frac{\pi}{4}\right) + \frac{1}{2} \times \frac{\pi}{4}\right) - \left(\frac{1}{4}\sin(2 \times 0) + \frac{1}{2} \times 0\right)$

$= \left(\frac{1}{4} + \frac{\pi}{8}\right) - (0 + 0) = \frac{1}{4} + \frac{\pi}{8}$

47)

$\int_0^1 \frac{1-x^2}{1+x^2} \, \mathrm{d}x$

$x = \tan u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u \, \mathrm{d}u$

Change limits:

$x = 1 \Rightarrow 1 = \tan u \Rightarrow u = \arctan 1 = \frac{\pi}{4}$

$x = 0 \Rightarrow 0 = \tan u \Rightarrow u = \arctan 0 = 0$

$$\int_0^1 \frac{1 - x^2}{1 + x^2} \, \mathrm{d}x = \int_0^{\frac{\pi}{4}} \frac{1 - \tan^2 u}{1 + \tan^2 u} \times \sec^2 u \, \mathrm{d}u$$
$$= \int_0^{\frac{\pi}{4}} \frac{1 - \tan^2 u}{\sec^2 u} \times \sec^2 u \, \mathrm{d}u = \int_0^{\frac{\pi}{4}} (1 - \tan^2 u) \, \mathrm{d}u$$
$$= \int_0^{\frac{\pi}{4}} (1 - (\sec^2 u - 1)) \, \mathrm{d}u = \int_0^{\frac{\pi}{4}} (2 - \sec^2 u) \, \mathrm{d}u$$

$= [2u - \tan u]_0^{\frac{\pi}{4}}$

$= \left(2 \times \frac{\pi}{4} - \tan\frac{\pi}{4}\right) - (2 \times 0 - \tan 0)$

$= \left(\frac{\pi}{2} - 1\right) - (0 - 0) = \frac{\pi}{2} - 1$

48 a)

$\int_2^3 \frac{1}{(x-2)^2+1}\,\mathrm{d}x$

$x = 2 + \tan u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}x = \sec^2 u\,\mathrm{d}u$

Change limits:

$x = 3 \Rightarrow 3 = 2 + \tan u \Rightarrow \tan u = 1$

$\Rightarrow u = \arctan 1 = \frac{\pi}{4}$

$x = 2 \Rightarrow 2 = 2 + \tan u \Rightarrow \tan u = 0$

$\Rightarrow u = \arctan 0 = 0$

$\int_2^3 \frac{1}{(x-2)^2+1}\,\mathrm{d}x = \int_0^{\frac{\pi}{4}} \frac{1}{(2+\tan u-2)^2+1} \times \sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{1}{\tan^2 u+1} \times \sec^2 u\,\mathrm{d}u = \int_0^{\frac{\pi}{4}} \frac{1}{\sec^2 u} \times \sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} 1\,\mathrm{d}u$

$= [u]_0^{\frac{\pi}{4}}$

$= \left(\frac{\pi}{4}\right) - (0) = \frac{\pi}{4}$

48 b)

$\int_{-3}^{-1} \frac{1}{(x+3)^2+4}\,\mathrm{d}x$

$x = 2\tan u - 3 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2\sec^2 u \Rightarrow \mathrm{d}x = 2\sec^2 u\,\mathrm{d}u$

Change limits:

$x = -1 \Rightarrow -1 = 2\tan u - 3 \Rightarrow \tan u = 1$

$\Rightarrow u = \arctan 1 = \frac{\pi}{4}$

$x = -3 \Rightarrow -3 = 2\tan u - 3 \Rightarrow \tan u = 0$

$\Rightarrow u = \arctan 0 = 0$

$\int_{-3}^{-1} \frac{1}{(x+3)^2+4}\,\mathrm{d}x = \int_0^{\frac{\pi}{4}} \frac{1}{(2\tan u-3+3)^2+4} \times 2\sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{1}{4\tan^2 u+4} \times 2\sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{1}{4(\tan^2 u+1)} \times 2\sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{1}{4\sec^2 u} \times 2\sec^2 u\,\mathrm{d}u = \int_0^{\frac{\pi}{4}} \frac{1}{2}\,\mathrm{d}u$

$= \left[\frac{1}{2}u\right]_0^{\frac{\pi}{4}}$

$= \left(\frac{1}{2} \times \frac{\pi}{4}\right) - \left(\frac{1}{2} \times 0\right) = \frac{\pi}{8}$

49)

$\int_5^8 \frac{1}{x^2-10x+34}\,\mathrm{d}x = \int_5^8 \frac{1}{(x-5)^2+9}\,\mathrm{d}x$

$x = 3\tan u + 5 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 3\sec^2 u \Rightarrow \mathrm{d}x = 3\sec^2 u\,\mathrm{d}u$

Change limits:

$x = 8 \Rightarrow 8 = 3\tan u + 5 \Rightarrow \tan u = 1$

$\Rightarrow u = \arctan 1 = \frac{\pi}{4}$

$x = 5 \Rightarrow 5 = 3\tan u + 5 \Rightarrow \tan u = 0$

$\Rightarrow u = \arctan 0 = 0$

$\int_5^8 \frac{1}{(x-5)^2+9}\,\mathrm{d}x = \int_0^{\frac{\pi}{4}} \frac{1}{(3\tan u+5-5)^2+9} \times 3\sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{1}{9\tan^2 u+9} \times 3\sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{1}{9(\tan^2 u+1)} \times 3\sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{1}{9\sec^2 u} \times 3\sec^2 u\,\mathrm{d}u = \int_0^{\frac{\pi}{4}} \frac{1}{3}\,\mathrm{d}u$

$= \left[\frac{1}{3}u\right]_0^{\frac{\pi}{4}}$

$= \left(\frac{1}{3} \times \frac{\pi}{4}\right) - \left(\frac{1}{3} \times 0\right) = \frac{\pi}{12}$

50)

$\int_5^8 \frac{12x}{x^2-10x+34}\,\mathrm{d}x = \int_5^8 \frac{12x}{(x-5)^2+9}\,\mathrm{d}x$

$x = 3\tan u + 5 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 3\sec^2 u \Rightarrow \mathrm{d}x = 3\sec^2 u\,\mathrm{d}u$

Change limits:

$x = 8 \Rightarrow 8 = 3\tan u + 5 \Rightarrow \tan u = 1$

$\Rightarrow u = \arctan 1 = \frac{\pi}{4}$

$x = 5 \Rightarrow 5 = 3\tan u + 5 \Rightarrow \tan u = 0$

$\Rightarrow u = \arctan 0 = 0$

$\int_5^8 \frac{12x}{(x-5)^2+9}\,\mathrm{d}x = \int_0^{\frac{\pi}{4}} \frac{12(3\tan u+5)}{(3\tan u+5-5)^2+9} \times 3\sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{36\tan u+60}{9\tan^2 u+9} \times 3\sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{36\tan u+60}{9(\tan^2 u+1)} \times 3\sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{36\tan u+60}{9\sec^2 u} \times 3\sec^2 u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \frac{1}{3}(36\tan u + 60)\,\mathrm{d}u = \int_0^{\frac{\pi}{4}} (12\tan u + 20)\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} \left(\frac{12\sin u}{\cos u} + 20\right)\,\mathrm{d}u$

$= [-12\ln|\cos u| + 20u]_0^{\frac{\pi}{4}}$

$= (-12\ln\left|\cos\tfrac{\pi}{4}\right| + 20 \times \tfrac{\pi}{4}) - (-12\ln|\cos 0| + 20 \times 0)$

$= \left(-12\ln\frac{\sqrt{2}}{2} + 5\pi\right) - (0+0) = -12\ln\frac{\sqrt{2}}{2} + 5\pi$

$= \ln\left(\frac{\sqrt{2}}{2}\right)^{-12} + 5\pi = \ln 64 + 5\pi$

51)

$\int_{\frac{2\sqrt{3}}{3}}^2 \frac{6}{x^2\sqrt{x^2-1}}\,\mathrm{d}x$

$x = \operatorname{cosec} u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = -\operatorname{cosec} u \cot u$

$\Rightarrow \mathrm{d}x = -\operatorname{cosec} u \cot u\,\mathrm{d}u$

Change limits:

$x = 2 \Rightarrow 2 = \operatorname{cosec} u \Rightarrow \sin u = \frac{1}{2} \Rightarrow u = \arcsin\frac{1}{2} = \frac{\pi}{6}$

$x = \frac{2\sqrt{3}}{3} \Rightarrow \frac{2\sqrt{3}}{3} = \operatorname{cosec} u \Rightarrow \sin u = \frac{3}{2\sqrt{3}} = \frac{\sqrt{3}}{2}$

$\Rightarrow u = \arcsin \frac{\sqrt{3}}{2} = \frac{\pi}{3}$

$$\begin{aligned}\int_{\frac{2\sqrt{3}}{3}}^{2} \frac{6}{x^2\sqrt{x^2-1}}\,\mathrm{d}x &= \int_{\frac{\pi}{3}}^{\frac{\pi}{6}} \frac{6}{\operatorname{cosec}^2 u\sqrt{\operatorname{cosec}^2 u - 1}} \times -\operatorname{cosec} u \cot u\,\mathrm{d}u \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{6}{\operatorname{cosec}^2 u\sqrt{\cot^2 u}} \times \operatorname{cosec} u \cot u\,\mathrm{d}u \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{6}{\operatorname{cosec}^2 u \cot u} \times \operatorname{cosec} u \cot u\,\mathrm{d}u \\ &= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{6}{\operatorname{cosec} u}\,\mathrm{d}u = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 6\sin u\,\mathrm{d}u\end{aligned}$$

$= [-6\cos u]_{\frac{\pi}{6}}^{\frac{\pi}{3}}$

$= \left(-6\cos\frac{\pi}{3}\right) - \left(-6\cos\frac{\pi}{6}\right)$

$= \left(-6 \times \frac{1}{2}\right) - \left(-6 \times \frac{\sqrt{3}}{2}\right) = (-3) - (-3\sqrt{3})$

$= 3\sqrt{3} - 3$

52)

$\int \frac{1}{x^2\sqrt{x^2-9}}\,\mathrm{d}x$

$x = 3\operatorname{cosec} u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = -3\operatorname{cosec} u \cot u$

$\Rightarrow \mathrm{d}x = -3\operatorname{cosec} u \cot u\,\mathrm{d}u$

$x = 3\operatorname{cosec} u \Rightarrow \operatorname{cosec} u = \frac{1}{3}x \Rightarrow \sin u = \frac{3}{x}$

$$\begin{aligned}\int \frac{1}{x^2\sqrt{x^2-9}}\,\mathrm{d}x &= \int \frac{1}{(3\operatorname{cosec} u)^2\sqrt{(3\operatorname{cosec} u)^2 - 9}} \times -3\operatorname{cosec} u \cot u\,\mathrm{d}u \\ &= \int \frac{1}{9\operatorname{cosec}^2 u\sqrt{9\operatorname{cosec}^2 u - 9}} \times -3\operatorname{cosec} u \cot u\,\mathrm{d}u \\ &= \int \frac{1}{9\operatorname{cosec}^2 u\sqrt{9(\operatorname{cosec}^2 u - 1)}} \times -3\operatorname{cosec} u \cot u\,\mathrm{d}u \\ &= \int \frac{1}{9\operatorname{cosec}^2 u\sqrt{9\cot^2 u}} \times -3\operatorname{cosec} u \cot u\,\mathrm{d}u \\ &= \int \frac{1}{9\operatorname{cosec}^2 u \times 3\cot u} \times -3\operatorname{cosec} u \cot u\,\mathrm{d}u\end{aligned}$$

$= \int -\frac{1}{9\operatorname{cosec} u}\,\mathrm{d}u = \int -\frac{1}{9}\sin u\,\mathrm{d}u = \frac{1}{9}\cos u + c$

$= \frac{1}{9}\sqrt{1 - \sin^2 u} + c = \frac{1}{9}\sqrt{1 - \left(\frac{3}{x}\right)^2} + c$

$= \frac{1}{9}\sqrt{1 - \frac{9}{x^2}} + c = \frac{1}{9}\sqrt{\frac{x^2-9}{x^2}} + c = \frac{1}{9} \times \frac{\sqrt{x^2-9}}{x} + c$

$= \frac{\sqrt{x^2-9}}{9x} + c$

53)

$\int (\ln 2x)^2\,\mathrm{d}x$

$2x = \mathrm{e}^u \Rightarrow x = \frac{1}{2}\mathrm{e}^u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \frac{1}{2}\mathrm{e}^u \Rightarrow \mathrm{d}x = \frac{1}{2}\mathrm{e}^u\,\mathrm{d}u$

$2x = \mathrm{e}^u \Rightarrow \ln 2x = u$

$\int (\ln 2x)^2 \,\mathrm{d}x = \int u^2 \times \frac{1}{2}\mathrm{e}^u \,\mathrm{d}u = \int \frac{1}{2}u^2\mathrm{e}^u \,\mathrm{d}u$

Now consider $\int \frac{1}{2}x^2\mathrm{e}^x \,\mathrm{d}x$ and use integration by parts:

$$\begin{array}{ll} u = \frac{1}{2}x^2 & \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^x \\ \frac{\mathrm{d}u}{\mathrm{d}x} = x & v = \mathrm{e}^x \end{array}$$

$\int \frac{1}{2}x^2\mathrm{e}^x \,\mathrm{d}x = \frac{1}{2}x^2\mathrm{e}^x - \int \mathrm{e}^x \times x \,\mathrm{d}x$

$= \frac{1}{2}x^2\mathrm{e}^x - \int x\mathrm{e}^x \,\mathrm{d}x$

$$\begin{array}{ll} u = x & \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^x \\ \frac{\mathrm{d}u}{\mathrm{d}x} = 1 & v = \mathrm{e}^x \end{array}$$

$\int \frac{1}{2}x^2\mathrm{e}^x \,\mathrm{d}x = \frac{1}{2}x^2\mathrm{e}^x - \int x\mathrm{e}^x \,\mathrm{d}x$

$= \frac{1}{2}x^2\mathrm{e}^x - (x\mathrm{e}^x - \int \mathrm{e}^x \,\mathrm{d}x)$

$= \frac{1}{2}x^2\mathrm{e}^x - x\mathrm{e}^x + \int \mathrm{e}^x \,\mathrm{d}x = \frac{1}{2}x^2\mathrm{e}^x - x\mathrm{e}^x + \mathrm{e}^x + c$

$\therefore \int (\ln 2x)^2 \,\mathrm{d}x = \int \frac{1}{2}u^2\mathrm{e}^u \,\mathrm{d}u$

$= \frac{1}{2}u^2\mathrm{e}^u - u\mathrm{e}^u + \mathrm{e}^u + c$

$= \frac{1}{2}(\ln 2x)^2\mathrm{e}^{\ln 2x} - (\ln 2x)\mathrm{e}^{\ln 2x} + \mathrm{e}^{\ln 2x} + c$

$= \frac{1}{2}(\ln 2x)^2 \times 2x - (\ln 2x) \times 2x + 2x + c$

$= x(\ln 2x)^2 - 2x \ln 2x + 2x + c$

54)

$\int \frac{3}{(1-x^2)^{\frac{5}{2}}} \,\mathrm{d}x$

$x = \sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u \,\mathrm{d}u$

$\int \frac{3}{(1-x^2)^{\frac{5}{2}}} \,\mathrm{d}x = \int \frac{3}{(1-\sin^2 u)^{\frac{5}{2}}} \times \cos u \,\mathrm{d}u$

$= \int \frac{3}{(\cos^2 u)^{\frac{5}{2}}} \times \cos u \,\mathrm{d}u = \int \frac{3}{\cos^5 u} \times \cos u \,\mathrm{d}u$

$= \int \frac{3}{\cos^4 u} \,\mathrm{d}u = \int 3\sec^4 u \,\mathrm{d}u$

$= \int 3\sec^2 u \times \sec^2 u \,\mathrm{d}u = \int 3\sec^2 u(\tan^2 u + 1) \,\mathrm{d}u$

Let $v = \tan u \Rightarrow \frac{\mathrm{d}v}{\mathrm{d}u} = \sec^2 u \Rightarrow \mathrm{d}u = \frac{1}{\sec^2 u}\mathrm{d}v$

$\int 3\sec^2 u(\tan^2 u + 1) \,\mathrm{d}u$

$= \int 3\sec^2 u(v^2 + 1) \times \frac{1}{\sec^2 u} \,\mathrm{d}v = \int 3(v^2 + 1) \,\mathrm{d}v$

$= \int (3v^2 + 3) \,\mathrm{d}v = v^3 + 3v + c$

$= \tan^3 u + 3\tan u + c$

$= \tan^3(\arcsin x) + 3\tan(\arcsin x) + c$

$\left(= \frac{x^3}{(1-x^2)^{\frac{3}{2}}} + \frac{3x}{\sqrt{1-x^2}} + c \text{ as } \tan(\arcsin x) = \frac{x}{\sqrt{1-x^2}}\right)$

55)

$\int \frac{4\sin 2x}{\cos x - 2} \,\mathrm{d}x$

$u = \cos x - 2 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = -\sin x \Rightarrow \mathrm{d}x = -\frac{1}{\sin x}\mathrm{d}u$

$u = \cos x - 2 \Rightarrow \cos x = u + 2$

$\int \frac{4\sin 2x}{\cos x - 2} \,\mathrm{d}x = \int \frac{8\sin x \cos x}{\cos x - 2} \,\mathrm{d}x$

$= \int \frac{8\sin x(u+2)}{u} \times -\frac{1}{\sin x}\,\mathrm{d}u = \int -\frac{8(u+2)}{u}\,\mathrm{d}u$

$= \int \frac{-8u-16}{u}\,\mathrm{d}u = \int \left(-\frac{8u}{u} - \frac{16}{u}\right)\,\mathrm{d}u = \int \left(-8 - \frac{16}{u}\right)\,\mathrm{d}u$

$= -8u - 16\ln|u| + c$

$= -8(\cos x - 2) - 16\ln|\cos x - 2| + c$

56)

$\int \frac{x^2}{\sqrt{1-x^2}}\,\mathrm{d}x$

$x = \sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = \cos u \Rightarrow \mathrm{d}x = \cos u\,\mathrm{d}u$

$\int \frac{x^2}{\sqrt{1-x^2}}\,\mathrm{d}x = \int \frac{\sin^2 u}{(1-\sin^2 u)^{\frac{1}{2}}} \times \cos u\,\mathrm{d}u$

$= \int \frac{\sin^2 u}{(\cos^2 u)^{\frac{1}{2}}} \times \cos u\,\mathrm{d}u = \int \frac{\sin^2 u}{\cos u} \times \cos u\,\mathrm{d}u$

$= \int \sin^2 u\,\mathrm{d}u$

$\cos 2u = 1 - 2\sin^2 u \Rightarrow \sin^2 u = \frac{1}{2}(1 - \cos 2u)$

$\therefore \int \sin^2 u\,\mathrm{d}u = \int \frac{1}{2}(1 - \cos 2u)\,\mathrm{d}u$

$= \int \left(\frac{1}{2} - \frac{1}{2}\cos 2u\right)\,\mathrm{d}u$

$= \frac{1}{2}u - \frac{1}{4}\sin 2u + c$

$= \frac{1}{2}u - \frac{1}{4} \times 2\sin u\cos u + c$

$= \frac{1}{2}u - \frac{1}{2}\sin u\sqrt{1-\sin^2 u} + c$

$= \frac{1}{2}\arcsin x - \frac{1}{2}x\sqrt{1-x^2} + c$

57 a)

$\int_0^{\frac{3}{2}} \sqrt{9-x^2}\,\mathrm{d}x$

$x = 3\sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 3\cos u \Rightarrow \mathrm{d}x = 3\cos u\,\mathrm{d}u$

Change limits:

$x = \frac{3}{2} \Rightarrow \frac{3}{2} = 3\sin u \Rightarrow \sin u = \frac{1}{2} \Rightarrow u = \arcsin\frac{1}{2} = \frac{\pi}{6}$

$x = 0 \Rightarrow 0 = 3\sin u \Rightarrow \sin u = 0 \Rightarrow u = 0$

$$\begin{aligned}\int_0^{\frac{3}{2}} \sqrt{9-x^2}\,\mathrm{d}x &= \int_0^{\frac{\pi}{6}} \sqrt{9-(3\sin u)^2} \times 3\cos u\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{6}} \sqrt{9-9\sin^2 u} \times 3\cos u\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{6}} \sqrt{9(1-\sin^2 u)} \times 3\cos u\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{6}} \sqrt{9\cos^2 u} \times 3\cos u\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{6}} 3\cos u \times 3\cos u\,\mathrm{d}u = \int_0^{\frac{\pi}{6}} 9\cos^2 u\,\mathrm{d}u\end{aligned}$$

$\cos 2u = 2\cos^2 u - 1 \Rightarrow \cos^2 u = \frac{1}{2}(\cos 2u + 1)$

$\therefore \int_0^{\frac{\pi}{6}} 9\cos^2 u\,\mathrm{d}u = \int_0^{\frac{\pi}{6}} 9 \times \frac{1}{2}(\cos 2u + 1)\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{6}} \left(\frac{9}{2}\cos 2u + \frac{9}{2}\right)\,\mathrm{d}u$

$= \left[\frac{9}{4}\sin 2u + \frac{9}{2}u\right]_0^{\frac{\pi}{6}}$

$= \left(\frac{9}{4}\sin\left(2\times\frac{\pi}{6}\right)+\frac{9}{2}\times\frac{\pi}{6}\right)-\left(\frac{9}{4}\sin(2\times 0)+\frac{9}{2}\times 0\right)$

$= \left(\frac{9}{4}\times\frac{\sqrt{3}}{2}+\frac{3\pi}{4}\right)-(0+0)=\frac{9\sqrt{3}}{8}+\frac{3\pi}{4}$

57 b)

$\int_0^{\frac{1}{3}}\sqrt{1-9x^2}\,\mathrm{d}x$

$x=\frac{1}{3}\sin u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u}=\frac{1}{3}\cos u \Rightarrow \mathrm{d}x=\frac{1}{3}\cos u\,\mathrm{d}u$

Change limits:

$x=\frac{1}{3}\Rightarrow\frac{1}{3}=\frac{1}{3}\sin u\Rightarrow\sin u=1\Rightarrow u=\arcsin 1=\frac{\pi}{2}$

$x=0\Rightarrow 0=\frac{1}{3}\sin u\Rightarrow\sin u=0\Rightarrow u=0$

$$\begin{aligned}\int_0^{\frac{1}{3}}\sqrt{1-9x^2}\,\mathrm{d}x &= \int_0^{\frac{\pi}{2}}\sqrt{1-9(\frac{1}{3}\sin u)^2}\times\frac{1}{3}\cos u\,\mathrm{d}u\\ &= \int_0^{\frac{\pi}{2}}\sqrt{1-\sin^2 u}\times\frac{1}{3}\cos u\,\mathrm{d}u\\ &= \int_0^{\frac{\pi}{2}}\sqrt{\cos^2 u}\times\frac{1}{3}\cos u\,\mathrm{d}u\\ &= \int_0^{\frac{\pi}{2}}\cos u\times\frac{1}{3}\cos u\,\mathrm{d}u=\int_0^{\frac{\pi}{2}}\frac{1}{3}\cos^2 u\,\mathrm{d}u\end{aligned}$$

$\cos 2u=2\cos^2 u-1\Rightarrow\cos^2 u=\frac{1}{2}(\cos 2u+1)$

$\therefore\int_0^{\frac{\pi}{2}}\frac{1}{3}\cos^2 u\,\mathrm{d}u=\int_0^{\frac{\pi}{2}}\frac{1}{3}\times\frac{1}{2}(\cos 2u+1)\,\mathrm{d}u$

$=\int_0^{\frac{\pi}{2}}\left(\frac{1}{6}\cos 2u+\frac{1}{6}\right)\,\mathrm{d}u$

$=\left[\frac{1}{12}\sin 2u+\frac{1}{6}u\right]_0^{\frac{\pi}{2}}$

$=\left(\frac{1}{12}\sin\left(2\times\frac{\pi}{2}\right)+\frac{1}{6}\times\frac{\pi}{2}\right)-\left(\frac{1}{12}\sin(2\times 0)+\frac{1}{6}\times 0\right)$

$=(0+\frac{\pi}{12})-(0+0)=\frac{\pi}{12}$

58 a)

$\int_0^1\frac{1-x^2}{(1+x^2)^2}\,\mathrm{d}x$

$x=\tan u\Rightarrow\frac{\mathrm{d}x}{\mathrm{d}u}=\sec^2 u\Rightarrow\mathrm{d}x=\sec^2 u\,\mathrm{d}u$

Change limits:

$x=1\Rightarrow 1=\tan u\Rightarrow u=\arctan 1=\frac{\pi}{4}$

$x=0\Rightarrow 0=\tan u\Rightarrow u=\arctan 0=0$

$$\begin{aligned}\int_0^1\frac{1-x^2}{(1+x^2)^2}\,\mathrm{d}x &= \int_0^{\frac{\pi}{4}}\frac{1-\tan^2 u}{(1+\tan^2 u)^2}\times\sec^2 u\,\mathrm{d}u\\ &= \int_0^{\frac{\pi}{4}}\frac{1-\tan^2 u}{(\sec^2 u)^2}\times\sec^2 u\,\mathrm{d}u=\int_0^{\frac{\pi}{4}}\frac{1-\tan^2 u}{\sec^2 u}\,\mathrm{d}u\\ &= \int_0^{\frac{\pi}{4}}\frac{1-(\sec^2 u-1)}{\sec^2 u}\,\mathrm{d}u=\int_0^{\frac{\pi}{4}}\frac{2-\sec^2 u}{\sec^2 u}\,\mathrm{d}u\\ &= \int_0^{\frac{\pi}{4}}\left(\frac{2}{\sec^2 u}-\frac{\sec^2 u}{\sec^2 u}\right)\,\mathrm{d}u=\int_0^{\frac{\pi}{4}}(2\cos^2 u-1)\,\mathrm{d}u\end{aligned}$$

$\cos 2u=2\cos^2 u-1\Rightarrow\cos^2 u=\frac{1}{2}(\cos 2u+1)$

$\therefore \int_0^{\frac{\pi}{4}} (2\cos^2 u - 1)\,\mathrm{d}u = \int_0^{\frac{\pi}{4}} \left(2 \times \frac{1}{2}(\cos 2u + 1) - 1\right)\,\mathrm{d}u = \int_0^{\frac{\pi}{4}} \cos 2u\,\mathrm{d}u$

$= \left[\frac{1}{2}\sin 2u\right]_0^{\frac{\pi}{4}}$

$= \left(\frac{1}{2}\sin\left(2 \times \frac{\pi}{4}\right)\right) - \left(\frac{1}{2}\sin(2 \times 0)\right)$

$= \left(\frac{1}{2}\right) - (0) = \frac{1}{2}$

58 b)

$\int_0^{\frac{5}{2}} \sqrt{\frac{x}{5-x}}\,\mathrm{d}x$

$x = 5\sin^2 u \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 10\sin u \cos u$

$\Rightarrow \mathrm{d}x = 10\sin u \cos u\,\mathrm{d}u$

Change limits:

$x = \frac{5}{2} \Rightarrow \frac{5}{2} = 5\sin^2 u \Rightarrow \sin^2 u = \frac{1}{2} \Rightarrow \sin u = \frac{1}{\sqrt{2}}$

$\Rightarrow u = \arcsin\frac{1}{\sqrt{2}} = \frac{\pi}{4}$

$x = 0 \Rightarrow 0 = 5\sin^2 u \Rightarrow \sin^2 u = 0 \Rightarrow \sin u = 0$

$\Rightarrow u = \arcsin 0 = 0$

$$\begin{aligned}\int_0^{\frac{5}{2}} \sqrt{\frac{x}{5-x}}\,\mathrm{d}x &= \int_0^{\frac{\pi}{4}} \sqrt{\frac{5\sin^2 u}{5 - 5\sin^2 u}} \times 10\sin u\cos u\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{4}} \sqrt{\frac{5\sin^2 u}{5(1-\sin^2 u)}} \times 10\sin u\cos u\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{4}} \sqrt{\frac{\sin^2 u}{\cos^2 u}} \times 10\sin u\cos u\,\mathrm{d}u \\ &= \int_0^{\frac{\pi}{4}} \frac{\sin u}{\cos u} \times 10\sin u\cos u\,\mathrm{d}u = \int_0^{\frac{\pi}{4}} 10\sin^2 u\,\mathrm{d}u\end{aligned}$$

$\cos 2u = 1 - 2\sin^2 u \Rightarrow \sin^2 u = \frac{1}{2}(1 - \cos 2u)$

$\therefore \int_0^{\frac{\pi}{4}} 10\sin^2 u\,\mathrm{d}u = \int_0^{\frac{\pi}{4}} 10 \times \frac{1}{2}(1 - \cos 2u)\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{4}} (5 - 5\cos 2u)\,\mathrm{d}u$

$= \left[5u - \frac{5}{2}\sin 2u\right]_0^{\frac{\pi}{4}}$

$= \left(5 \times \frac{\pi}{4} - \frac{5}{2}\sin\left(2 \times \frac{\pi}{4}\right)\right) - \left(5 \times 0 - \frac{5}{2}\sin(2 \times 0)\right)$

$= \left(\frac{5\pi}{4} - \frac{5}{2}\right) - (0 - 0) = \frac{5\pi}{4} - \frac{5}{2}$

59 a)

$\int 2x^3 \ln(x^2 + 4)\,\mathrm{d}x$

$u = x^2 + 4 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = 2x \Rightarrow \mathrm{d}x = \frac{1}{2x}\mathrm{d}u$

$u = x^2 + 4 \Rightarrow x^2 = u - 4$

$\int 2x^3 \ln(x^2 + 4)\,\mathrm{d}x = \int 2x^3 \times \ln u \times \frac{1}{2x}\,\mathrm{d}u = \int x^2 \times \ln u\,\mathrm{d}u = \int (u - 4)\ln u\,\mathrm{d}u$

Now consider $\int (x - 4)\ln x\,\mathrm{d}x$ and use integration by parts:

$$\begin{array}{ll} u = \ln x & \frac{\mathrm{d}v}{\mathrm{d}x} = x - 4 \\ \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} & v = \frac{1}{2}x^2 - 4x \end{array}$$

$\int (x-4)\ln x\,\mathrm{d}x$

$= \left(\frac{1}{2}x^2 - 4x\right)\ln x - \int \left(\frac{1}{2}x^2 - 4x\right) \times \frac{1}{x}\,\mathrm{d}x$

$= \left(\frac{1}{2}x^2 - 4x\right)\ln x - \int \left(\frac{1}{2}x - 4\right)\,\mathrm{d}x$

$= \left(\frac{1}{2}x^2 - 4x\right)\ln x - \left(\frac{1}{4}x^2 - 4x\right) + c$

$= \left(\frac{1}{2}x^2 - 4x\right)\ln x - \frac{1}{4}x^2 + 4x + c$

$\therefore \int 2x^3 \ln(x^2+4)\,\mathrm{d}x = \int (u-4)\ln u\,\mathrm{d}u$

$= \left(\frac{1}{2}u^2 - 4u\right)\ln u - \frac{1}{4}u^2 + 4u + c$

$$= \left(\frac{1}{2}(x^2+4)^2 - 4(x^2+4)\right)\ln(x^2+4) - \frac{1}{4}(x^2+4)^2 + 4(x^2+4) + c$$

59 b)

$\int 3^x \ln(3^x+1)\,\mathrm{d}x$

$u = 3^x + 1 \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = (\ln 3)3^x \Rightarrow \mathrm{d}x = \frac{1}{(\ln 3)3^x}\mathrm{d}u$

$\int 3^x \ln(3^x+1)\,\mathrm{d}x = \int 3^x \times \ln u \times \frac{1}{(\ln 3)3^x}\,\mathrm{d}u$

$= \int \ln u \times \frac{1}{\ln 3}\,\mathrm{d}u$

Now consider $\frac{1}{\ln 3}\int \ln x\,\mathrm{d}x$ and use integration by parts:

$$\begin{array}{ll} u = \ln x & \frac{\mathrm{d}v}{\mathrm{d}x} = 1 \\ \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{x} & v = x \end{array}$$

$\frac{1}{\ln 3}\int \ln x\,\mathrm{d}x$

$= \frac{1}{\ln 3}\left(x\ln x - \int x \times \frac{1}{x}\,\mathrm{d}x\right) = \frac{1}{\ln 3}\left(x\ln x - \int 1\,\mathrm{d}x\right)$

$= \frac{1}{\ln 3}(x\ln x - x) + c$

$\therefore \int 3^x \ln(3^x+1)\,\mathrm{d}x = \int \ln u \times \frac{1}{\ln 3}\,\mathrm{d}u$

$= \frac{1}{\ln 3}(u\ln u - u) + c$

$= \frac{1}{\ln 3}\left((3^x+1)\ln(3^x+1) - (3^x+1)\right) + c$

60)

$\int \mathrm{e}^{4\sqrt{x+2}}\,\mathrm{d}x$

$u = (x+2)^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}(x+2)^{-\frac{1}{2}}$

$\Rightarrow \mathrm{d}x = 2(x+2)^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

$\int \mathrm{e}^{4\sqrt{x+2}}\,\mathrm{d}x = \int \mathrm{e}^{4u} \times 2u\,\mathrm{d}u = \int 2u\mathrm{e}^{4u}\,\mathrm{d}u$

Now consider $\int 2x\mathrm{e}^{4x}\,\mathrm{d}x$ and use integration by parts:

$$\begin{array}{ll} u = 2x & \frac{\mathrm{d}v}{\mathrm{d}x} = \mathrm{e}^{4x} \\ \frac{\mathrm{d}u}{\mathrm{d}x} = 2 & v = \frac{1}{4}\mathrm{e}^{x} \end{array}$$

$\int 2x\mathrm{e}^{4x}\,\mathrm{d}x = \frac{1}{2}x\mathrm{e}^{4x} - \int \frac{1}{4}\mathrm{e}^{4x} \times 2\,\mathrm{d}x$

$= \frac{1}{2}xe^{4x} - \int \frac{1}{2}e^{4x}\,dx = \frac{1}{2}xe^{4x} - \frac{1}{8}e^{4x} + c$

$\therefore \int e^{4\sqrt{x+2}}\,dx = \int 2ue^{4u}\,du = \frac{1}{2}ue^{4u} - \frac{1}{8}e^{4u} + c$

$= \frac{1}{2}\sqrt{x+2}\,e^{4\sqrt{x+2}} - \frac{1}{8}e^{4\sqrt{x+2}} + c$

61)

$\int_0^4 e^{\sqrt{2x+1}}\,dx$

$u^2 = 2x+1 \Rightarrow u = (2x+1)^{\frac{1}{2}} \Rightarrow \frac{du}{dx} = (2x+1)^{-\frac{1}{2}}$

$\Rightarrow dx = (2x+1)^{\frac{1}{2}}du \Rightarrow dx = u\,du$

Change limits:

$x = 4 \Rightarrow u^2 = 9 \Rightarrow u = 3$

$x = 0 \Rightarrow u^2 = 1 \Rightarrow u = 1$

$\int_0^4 e^{\sqrt{2x+1}}\,dx = \int_1^3 e^u \times u\,du = \int_1^3 ue^u\,du$

Now consider $\int xe^x\,dx$ and use integration by parts:

$$\begin{array}{ll} u = x & \frac{dv}{dx} = e^x \\ \frac{du}{dx} = 1 & v = e^x \end{array}$$

$\int xe^x\,dx = xe^x - \int e^x \times 1\,dx$

$= xe^x - \int e^x\,dx = xe^x - e^x + c$

$\therefore \int_0^4 e^{\sqrt{2x+1}}\,dx = \int_1^3 ue^u\,du = [ue^u - e^u]_1^3$

$= (3 \times e^3 - e^3) - (1 \times e^1 - e^1) = (2e^3) - (0) = 2e^3$

62 a)

$\int_0^1 5x^5 \ln(10x^3+1)\,dx$

$u = 10x^3 + 1 \Rightarrow \frac{du}{dx} = 30x^2 \Rightarrow dx = \frac{1}{30x^2}du$

$u = 10x^3 + 1 \Rightarrow x^3 = \frac{u-1}{10}$

Change limits:

$x = 1 \Rightarrow u = 11$

$x = 0 \Rightarrow u = 1$

$$\begin{aligned} \int_0^1 5x^5 \ln(10x^3+1)\,dx &= \int_1^{11} 5x^5 \times \ln u \times \frac{1}{30x^2}\,du \\ &= \int_1^{11} \frac{1}{6}x^3 \times \ln u\,du \\ &= \int_1^{11} \frac{1}{6}\left(\frac{u-1}{10}\right) \times \ln u\,du = \int_1^{11} \frac{1}{60}(u-1)\ln u\,du \end{aligned}$$

Now consider $\frac{1}{60}\int (x-1)\ln x\,dx$ and use integration by parts:

$$\begin{array}{ll} u = \ln x & \frac{dv}{dx} = x - 1 \\ \frac{du}{dx} = \frac{1}{x} & v = \frac{1}{2}x^2 - x \end{array}$$

$\frac{1}{60}\int (x-1)\ln x\,dx$

$= \frac{1}{60}\left(\left(\frac{1}{2}x^2 - x\right)\ln x - \int \left(\frac{1}{2}x^2 - x\right) \times \frac{1}{x}\,dx\right)$

$= \frac{1}{60}\left(\left(\frac{1}{2}x^2 - x\right)\ln x - \int\left(\frac{1}{2}x - 1\right)\,\mathrm{d}x\right)$

$= \frac{1}{60}\left(\left(\frac{1}{2}x^2 - x\right)\ln x - \left(\frac{1}{4}x^2 - x\right)\right) + c$

$= \frac{1}{60}\left(\frac{1}{2}x^2 - x\right)\ln x - \frac{1}{240}x^2 + \frac{1}{60}x + c$

$\therefore \int_0^1 5x^5\ln(10x^3+1)\,\mathrm{d}x = \int_1^{11}\frac{1}{60}(u-1)\ln u\,\mathrm{d}u$

$= \left[\frac{1}{60}\left(\frac{1}{2}u^2 - u\right)\ln u - \frac{1}{240}u^2 + \frac{1}{60}u\right]_1^{11}$

$$= \left(\frac{1}{60}\left(\frac{1}{2}\times 11^2 - 11\right)\ln 11 - \frac{1}{240}\times 11^2 + \frac{1}{60}\times 11\right)$$
$$- \left(\frac{1}{60}\left(\frac{1}{2}\times 1^2 - 1\right)\ln 1 - \frac{1}{240}\times 1^2 + \frac{1}{60}\times 1\right)$$

$= \left(\frac{33}{40}\ln 11 - \frac{121}{240} + \frac{11}{60}\right) - \left(0 - \frac{1}{240} + \frac{1}{60}\right)$

$= \left(\frac{33}{40}\ln 11 - \frac{77}{240}\right) - \left(\frac{1}{80}\right) = \frac{33}{40}\ln 11 - \frac{1}{3}$

62 b)

$\int_{-1}^{\frac{\pi^2}{4}-1}\cos(\sqrt{x+1})\,\mathrm{d}x$

$u = (x+1)^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}(x+1)^{-\frac{1}{2}}$

$\Rightarrow \mathrm{d}x = 2(x+1)^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

Change limits:

$x = \frac{\pi^2}{4} - 1 \Rightarrow u = \left(\frac{\pi^2}{4} - 1 + 1\right)^{\frac{1}{2}} = \left(\frac{\pi^2}{4}\right)^{\frac{1}{2}} = \frac{\pi}{2}$

$x = -1 \Rightarrow u = (-1+1)^{\frac{1}{2}} = 0$

$\int_{-1}^{\frac{\pi^2}{4}-1}\cos(\sqrt{x+1})\,\mathrm{d}x = \int_0^{\frac{\pi}{2}}\cos u \times 2u\,\mathrm{d}u$

$= \int_0^{\frac{\pi}{2}} 2u\cos u\,\mathrm{d}u$

Now consider $\int_0^{\frac{\pi}{2}} 2x\cos x\,\mathrm{d}x$ and use integration by parts:

$$\begin{array}{ll} u = 2x & \frac{\mathrm{d}v}{\mathrm{d}x} = \cos x \\ \frac{\mathrm{d}u}{\mathrm{d}x} = 2 & v = \sin x \end{array}$$

$\int_0^{\frac{\pi}{2}} 2x\cos x\,\mathrm{d}x = [2x\sin x]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 2\sin x\,\mathrm{d}x$

$= [2x\sin x]_0^{\frac{\pi}{2}} - [-2\cos x]_0^{\frac{\pi}{2}}$

$= [2x\sin x + 2\cos x]_0^{\frac{\pi}{2}}$

$= (2\times\frac{\pi}{2}\times\sin\frac{\pi}{2} + 2\cos\frac{\pi}{2}) - (2\times 0\times\sin 0 + 2\cos 0) = (\pi + 0) - (0+2) = \pi - 2$

63)

$\int_0^{\ln 5}\frac{10}{5+\mathrm{e}^x}\,\mathrm{d}x$

$u = \mathrm{e}^x \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \mathrm{e}^x \Rightarrow \mathrm{d}x = \frac{1}{\mathrm{e}^x}\mathrm{d}u \Rightarrow \mathrm{d}x = \frac{1}{u}\mathrm{d}u$

Change limits:

$x = \ln 5 \Rightarrow u = \mathrm{e}^{\ln 5} = 5$

$x = 0 \Rightarrow u = \mathrm{e}^0 = 1$

$\int_0^{\ln 5} \frac{10}{5+e^x}\,\mathrm{d}x = \int_1^5 \frac{10}{5+u} \times \frac{1}{u}\,\mathrm{d}u = \int_1^5 \frac{10}{(5+u)u}\,\mathrm{d}u$

$\frac{10}{(5+u)u} \equiv \frac{A}{5+u} + \frac{B}{u} \Rightarrow 10 \equiv Au + B(5+u)$

Let $u = -5$: $10 = -5A \Rightarrow A = -2$

Let $u = 0$: $10 = 5B \Rightarrow B = 2$

$\therefore \int_1^5 \frac{10}{(5+u)u}\,\mathrm{d}u = \int_1^5 \left(-\frac{2}{5+u} + \frac{2}{u}\right)\,\mathrm{d}u$

$= [-2\ln|5+u| + 2\ln|u|]_1^5$

$= (-2\ln|5+5| + 2\ln|5|) - (-2\ln|5+1| + 2\ln|1|)$

$= (-2\ln 10 + 2\ln 5) - (-2\ln 6 + 2\ln 1)$

$= (2(\ln 5 - \ln 10)) - (-2\ln 6 + 0)$

$= 2\ln\frac{5}{10} + 2\ln 6 = 2\left(\ln\frac{1}{2} + \ln 6\right) = 2\ln 3 = \ln 9$

64)

$\int_1^4 \frac{2}{x(1+3\sqrt{x})}\,\mathrm{d}x$

$x = u^2 \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}u} = 2u \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

$x = u^2 \Rightarrow \sqrt{x} = u$

Change limits:

$x = 4 \Rightarrow 4 = u^2 \Rightarrow u = 2$

$x = 1 \Rightarrow 1 = u^2 \Rightarrow u = 1$

$\int_1^4 \frac{2}{x(1+3\sqrt{x})}\,\mathrm{d}x = \int_1^2 \frac{2}{u^2(1+3u)} \times 2u\,\mathrm{d}u = \int_1^2 \frac{4}{u(1+3u)}\,\mathrm{d}u$

$\frac{4}{u(1+3u)} \equiv \frac{A}{u} + \frac{B}{1+3u} \Rightarrow 4 \equiv A(1+3u) + Bu$

Let $u = 0$: $4 = A$

Let $u = -\frac{1}{3}$: $4 = -\frac{1}{3}B \Rightarrow B = -12$

$\therefore \int_1^2 \frac{4}{u(1+3u)}\,\mathrm{d}u = \int_1^2 \left(\frac{4}{u} - \frac{12}{1+3u}\right)\,\mathrm{d}u$

$= [4\ln|u| - 4\ln|1+3u|]_1^2$

$= (4\ln|2| - 4\ln|1+3\times 2|) - (4\ln|1| - 4\ln|1+3\times 1|)$

$= (4\ln 2 - 4\ln 7) - (4\ln 1 - 4\ln 4)$

$= (4(\ln 2 - \ln 7)) - (0 - 4\ln 4) = 4\ln\frac{2}{7} + 4\ln 4$

$= 4\left(\ln\frac{2}{7} + \ln 4\right) = 4\ln\frac{8}{7}$

65 a)

$\int_1^3 \frac{1}{x\sqrt{x+1}}\,\mathrm{d}x$

$u = (x+1)^{\frac{1}{2}} \Rightarrow \frac{\mathrm{d}u}{\mathrm{d}x} = \frac{1}{2}(x+1)^{-\frac{1}{2}}$

$\Rightarrow \mathrm{d}x = 2(x+1)^{\frac{1}{2}}\mathrm{d}u \Rightarrow \mathrm{d}x = 2u\,\mathrm{d}u$

$u = (x+1)^{\frac{1}{2}} \Rightarrow u^2 = x+1 \Rightarrow x = u^2 - 1$

Change limits:

$x = 3 \Rightarrow u = \sqrt{3+1} = 2$

$x = 1 \Rightarrow u = \sqrt{1+1} = \sqrt{2}$

$\int_1^3 \frac{1}{x\sqrt{x+1}}\,\mathrm{d}x = \int_{\sqrt{2}}^2 \frac{1}{(u^2-1)u} \times 2u\,\mathrm{d}u = \int_{\sqrt{2}}^2 \frac{2}{(u-1)(u+1)}\,\mathrm{d}u$

$\frac{2}{(u-1)(u+1)} \equiv \frac{A}{u-1} + \frac{B}{u+1} \Rightarrow 2 \equiv A(u+1) + B(u-1)$

Let $u = 1$: $2 = 2A \Rightarrow A = 1$

Let $u = -1$: $2 = -2B \Rightarrow B = -1$

$\therefore \int_{\sqrt{2}}^{2} \frac{2}{(u-1)(u+1)}\,\mathrm{d}u = \int_{\sqrt{2}}^{2} \left(\frac{1}{u-1} - \frac{1}{u+1}\right)\,\mathrm{d}u$

$= [\ln|u-1| - \ln|u+1|]_{\sqrt{2}}^{2}$

$= (\ln|2-1| - \ln|2+1|) - (\ln|\sqrt{2}-1| - \ln|\sqrt{2}+1|)$

$= (\ln 1 - \ln 3) - (\ln(\sqrt{2}-1) - \ln(\sqrt{2}+1))$

$= (0 - \ln 3) - \ln\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right) = -\ln 3 - \ln\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)$

$= -\left(\ln 3 + \ln\left(\frac{\sqrt{2}-1}{\sqrt{2}+1}\right)\right) = -\ln\left(\frac{3\sqrt{2}-3}{\sqrt{2}+1}\right) = \ln\left(\frac{\sqrt{2}+1}{3\sqrt{2}-3}\right)$

$= \ln\left(\frac{3+2\sqrt{2}}{3}\right)$

65 b)

When $x = 1$, $y = \frac{1}{1\times\sqrt{1+1}} = \frac{1}{\sqrt{2}}$

When $x = 3$, $y = \frac{1}{3\times\sqrt{3+1}} = \frac{1}{6}$

Area of Trapezium $= \frac{1}{2} \times 2 \times \left(\frac{1}{\sqrt{2}} + \frac{1}{6}\right) = \frac{1+3\sqrt{2}}{6}$

$\therefore$ Area $= \frac{1+3\sqrt{2}}{6} - \ln\left(\frac{3+2\sqrt{2}}{3}\right)$

66 a)

$\int \sqrt{\mathrm{e}^x - 1}\,\mathrm{d}x$

$\mathrm{e}^x = \sec^2 u \Rightarrow \mathrm{e}^x \frac{\mathrm{d}x}{\mathrm{d}u} = 2\sec^2 u \tan u$

$\Rightarrow \mathrm{d}x = \frac{2\sec^2 u \tan u}{\mathrm{e}^x}\mathrm{d}u \Rightarrow \mathrm{d}x = \frac{2\sec^2 u \tan u}{\sec^2 u}\mathrm{d}u$

$\Rightarrow \mathrm{d}x = 2\tan u\,\mathrm{d}u$

$\int \sqrt{\mathrm{e}^x - 1}\,\mathrm{d}x = \int \sqrt{\sec^2 u - 1} \times 2\tan u\,\mathrm{d}u$

$= \int \sqrt{\tan^2 u} \times 2\tan u\,\mathrm{d}u = \int \tan u \times 2\tan u\,\mathrm{d}u$

$= \int 2\tan^2 u\,\mathrm{d}u$

66 b)

$\int_0^{\ln \frac{4}{3}} \sqrt{e^x - 1}\, du$

Change limits:

$e^{\ln(\frac{4}{3})} = \sec^2 u \Rightarrow \frac{4}{3} = \sec^2 u \Rightarrow \sec u = \frac{2}{\sqrt{3}}$

$\Rightarrow \cos u = \frac{\sqrt{3}}{2} \Rightarrow u = \arccos \frac{\sqrt{3}}{2} = \frac{\pi}{6}$

$e^0 = \sec^2 u \Rightarrow \sec u = 1 \Rightarrow \cos u = 1$

$\Rightarrow u = \arccos 1 = 0$

$\int_0^{\ln \frac{4}{3}} \sqrt{e^x - 1}\, du = \int_0^{\frac{\pi}{6}} 2\tan^2 u\, du$

$= \int_0^{\frac{\pi}{6}} 2(\sec^2 u - 1)\, du = \int_0^{\frac{\pi}{6}} (2\sec^2 u - 2)\, du$

$= [2\tan u - 2u]_0^{\frac{\pi}{6}}$

$= (2\tan \frac{\pi}{6} - 2 \times \frac{\pi}{6}) - (2\tan 0 - 2 \times 0)$

$= \left(2 \times \frac{\sqrt{3}}{3} - \frac{\pi}{3}\right) - (0 - 0) = \frac{2\sqrt{3}}{3} - \frac{\pi}{3}$